

CDS505

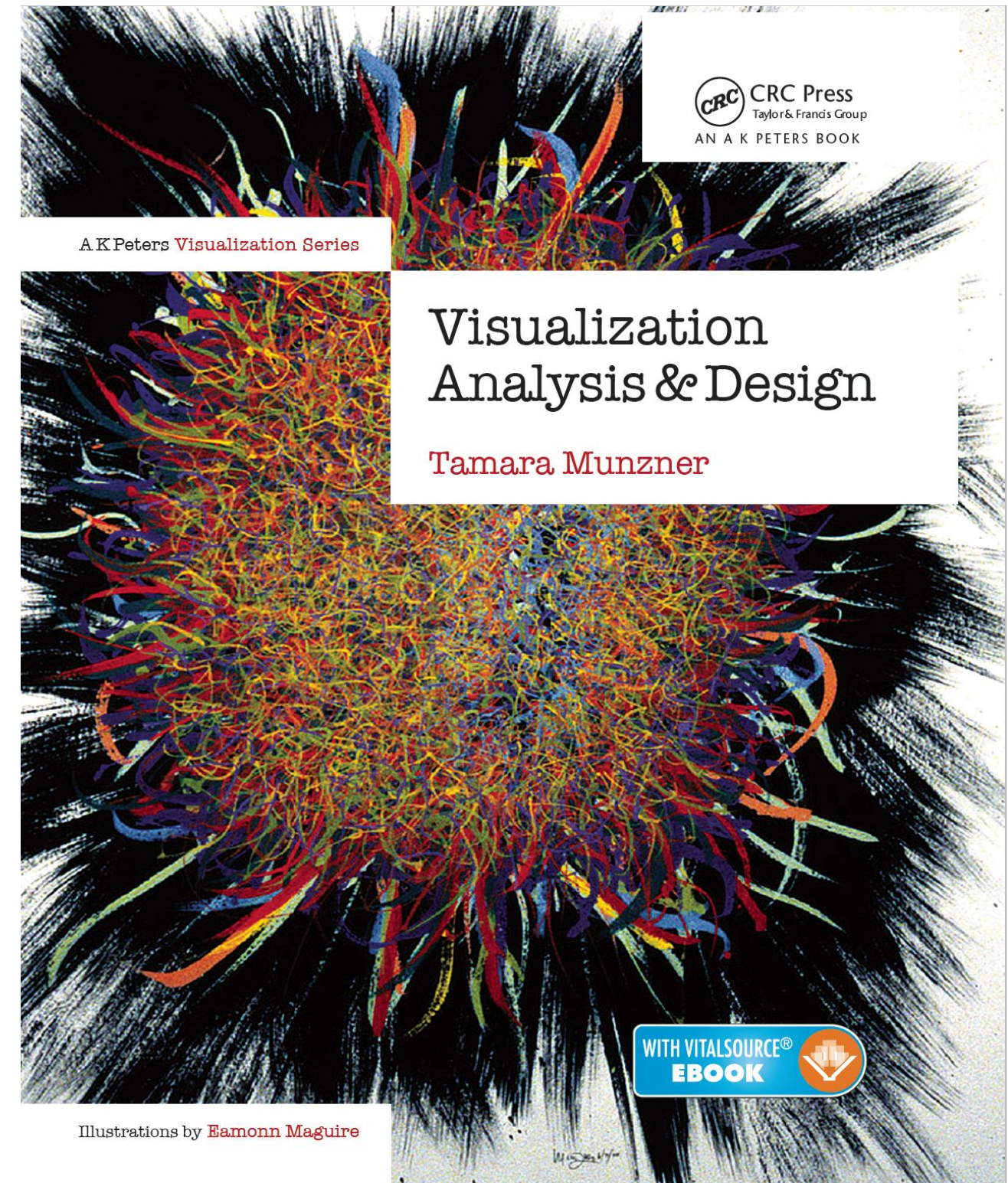
Data Visualization & Visual Analytics

Marks & Channels

*Nur Intan Raihana Ruhaiyem, PhD.
School of Computer Sciences
Universiti Sains Malaysia*

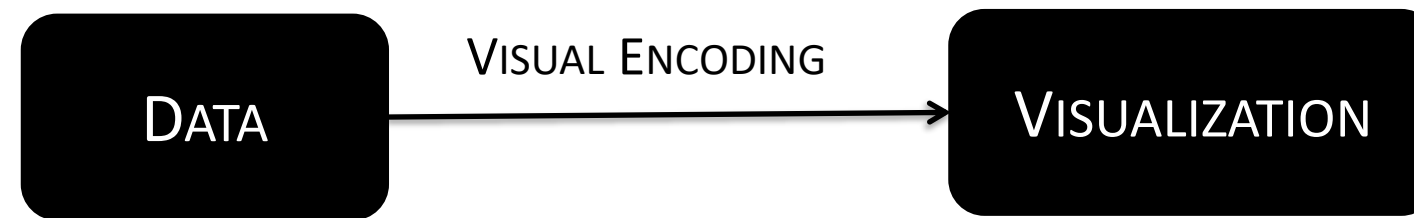
Website:

<https://cs.usm.my/index.php/faculty-member/194-nur-intan-raihana-ruhaiyem-dr>



- ✓ Recap previous class
- ✓ Marks and Channels
- ✓ Class activity

What is visual encoding?



Visual Encoding is the way in which data is mapped to visual structures. Typically, by mapping *data items* to visual *marks* and *data attributes* to visual *channels* in a *principled* way.

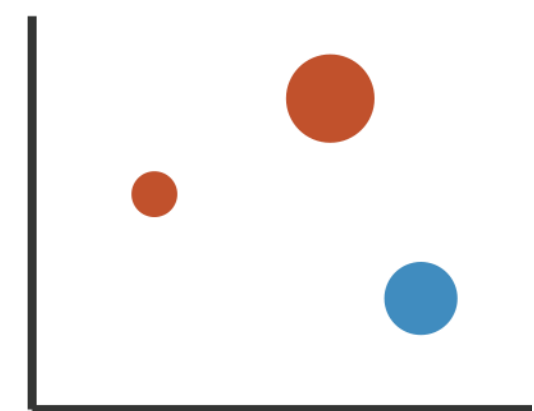
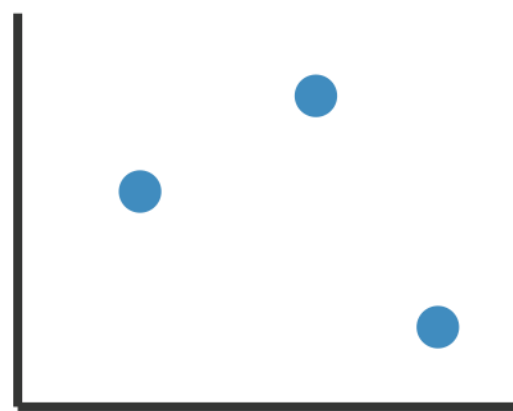
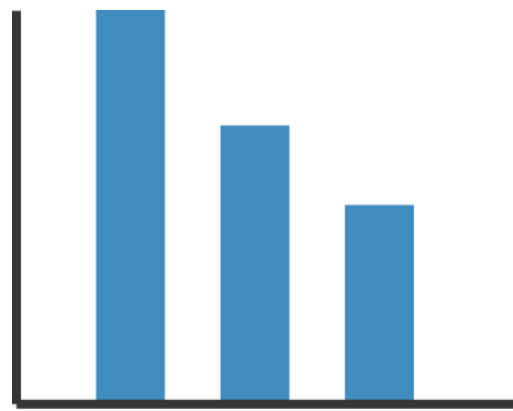
Every visualization can be described as a set of *mappings*:

From **data items** to visual **marks**.

From **data attributes** to visual **channels**.

Marks and **Attributes** – are the *alphabets* of visual encoding

- How to systematically analyze idiom structure?



- Marks & Channels
- Marks: represent items or links
- Channels: change appearance of marks based on attributes

- Basic geometric elements

➞ Points



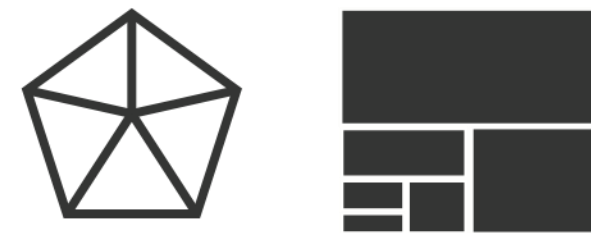
0D

➞ Lines



1D

➞ Areas

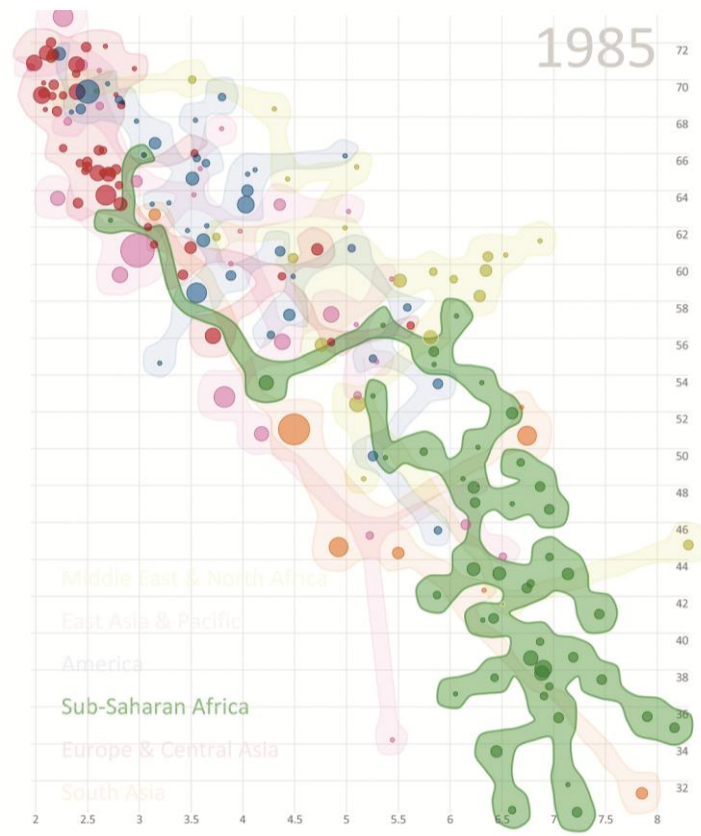
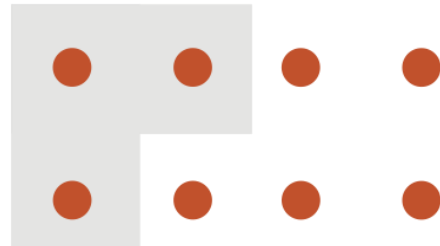


2D

- 3D mark: volume, rarely used

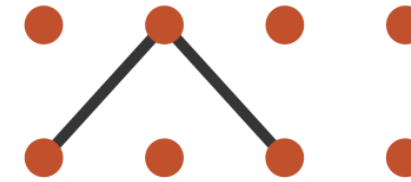
Marks for links

➔ Containment



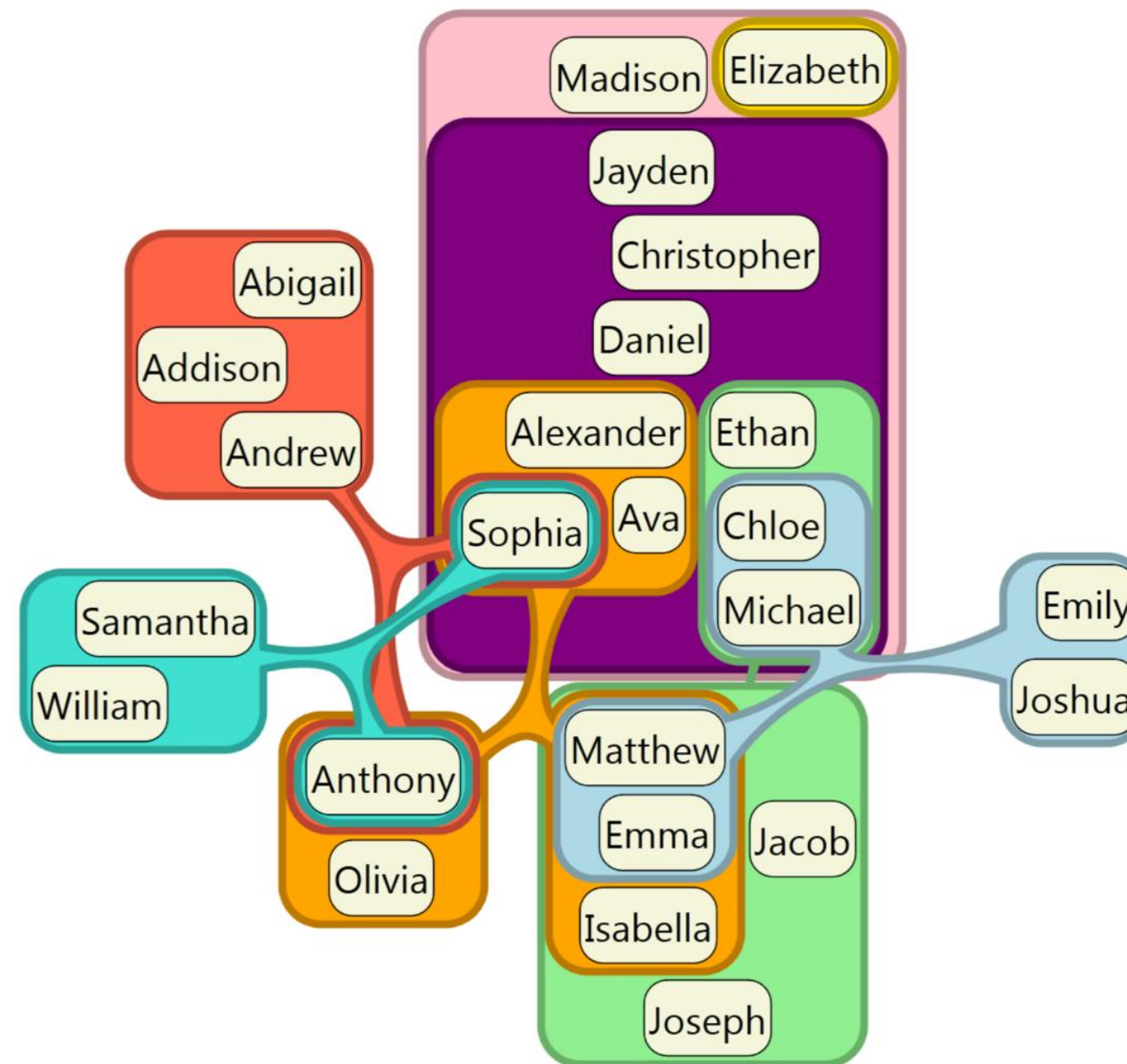
vialab.science.uoit.ca/portfolio/bubblesets

➔ Connection



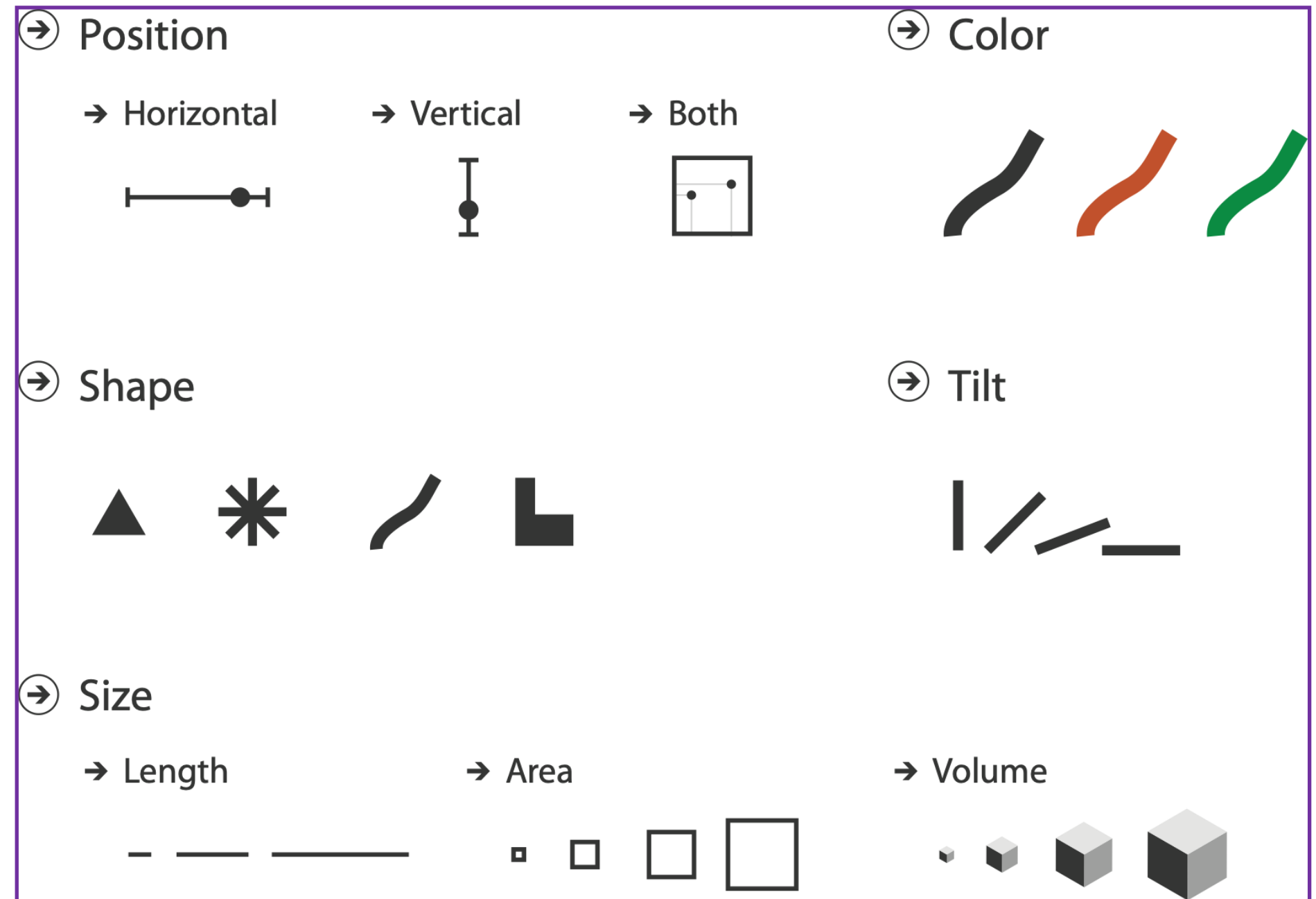
<https://observablehq.com/@d3/force-directed-graph>

Containment can be nested



https://www.microsoft.com/en-us/research/wp-content/uploads/2016/12/eulerdiagrams_infovis2010.pdf

- Control appearance of marks
- Proportional to or based on attributes
- Many names:
 - visual channels
 - visual variables
 - retinal channels
 - visual dimensions
 - ...



Definitions: Marks and Channels

- Marks
 - geometric primitives
- Channels
 - control appearance of marks
- Channel properties differ
- Type & amount of information that can be conveyed to human perceptual system

➞ Points



➞ Lines



➞ Interlocking Areas



➞ Position

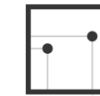
➞ Horizontal



➞ Vertical



➞ Both



➞ Color



➞ Shape



➞ Tilt



➞ Size

➞ Length



➞ Area



➞ Volume



Marks are the basic visual objects/units that represent **data items** visually (geometric primitives).

➞ Points



➞ Lines

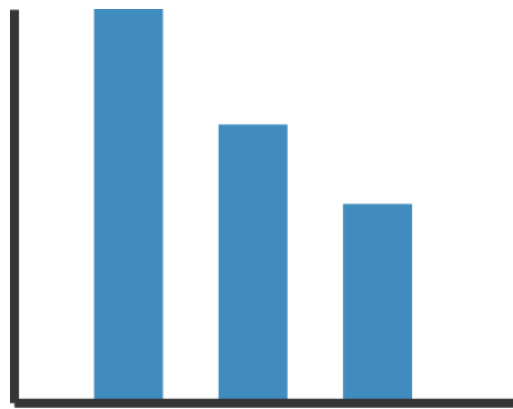


➞ Areas



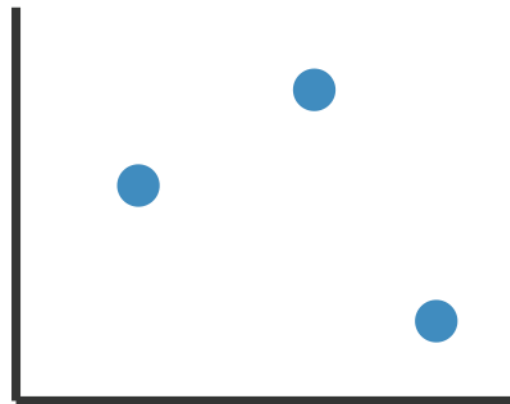
Visual channels are visual variables or properties we can use to represent characteristics of these data items i.e. **attributes**.

- Analyze idiom structure as combination of marks and channels



1:
vertical position

mark: line



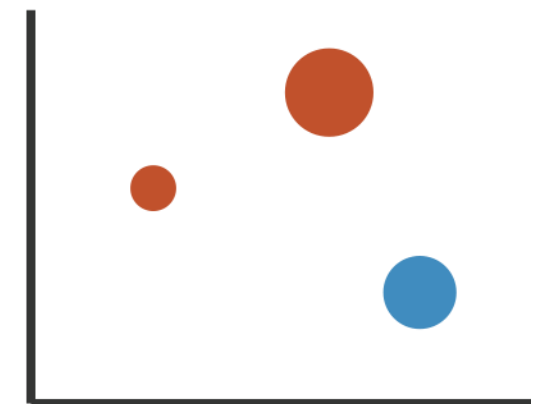
2:
vertical position
horizontal position

mark: point



3:
vertical position
horizontal position
color hue

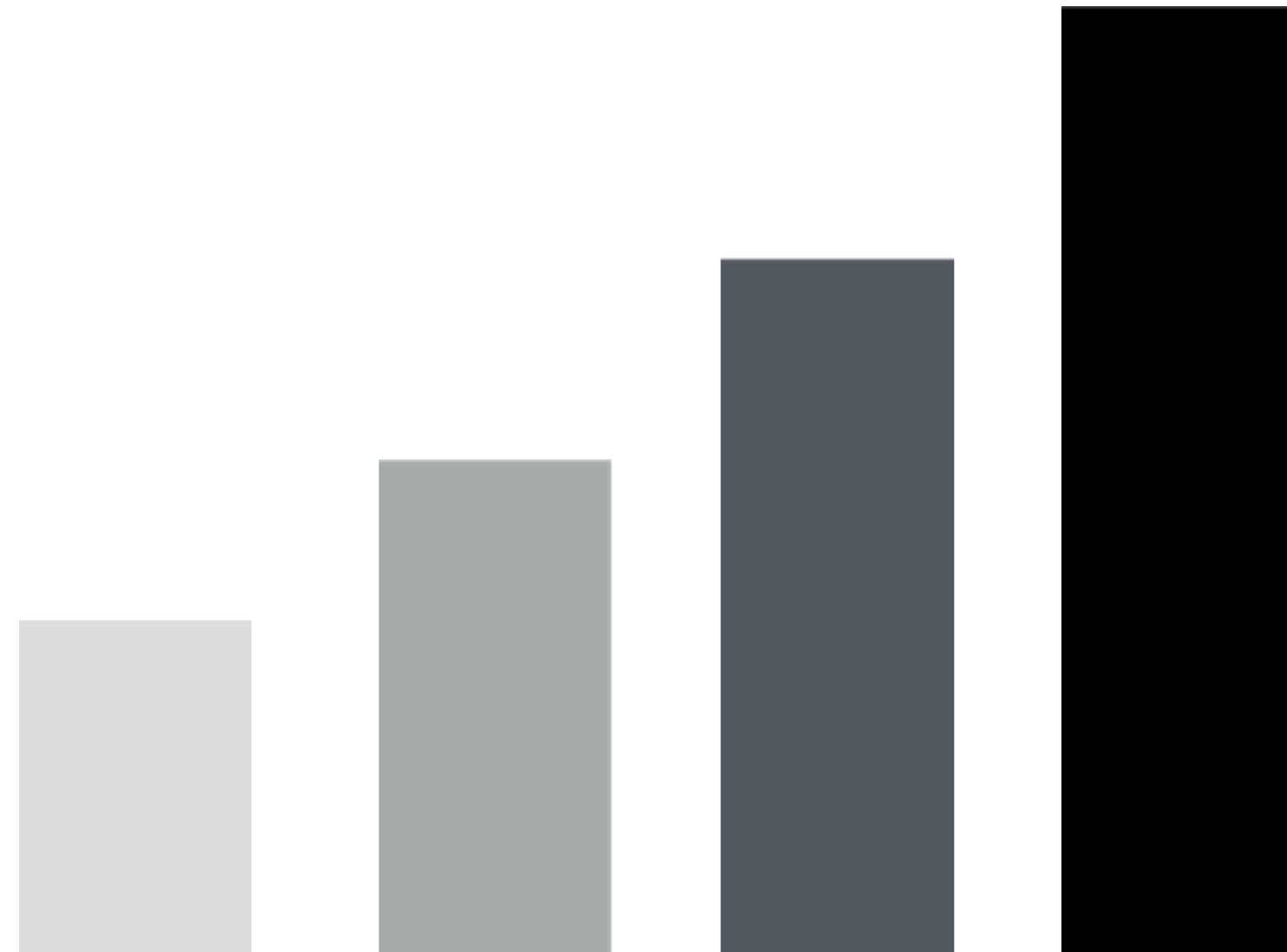
mark: point



4:
vertical position
horizontal position
color hue
size (area)

mark: point

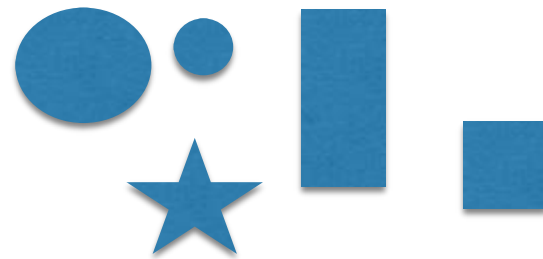
- Multiple channels
- Sends stronger message
... but uses up channels



Length and Luminance

- Math view: geometric primitives have dimensions

➞ Points



➞ Lines



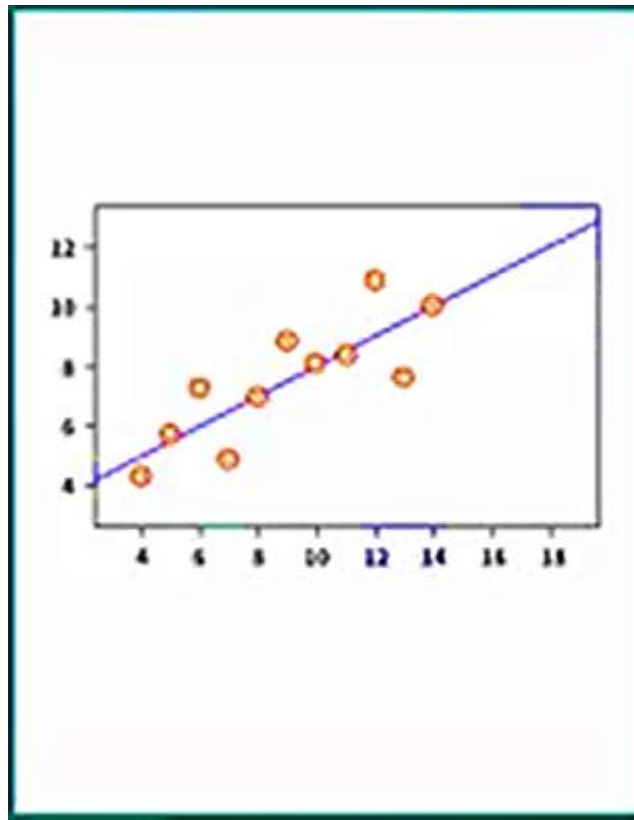
➞ Interlocking Areas



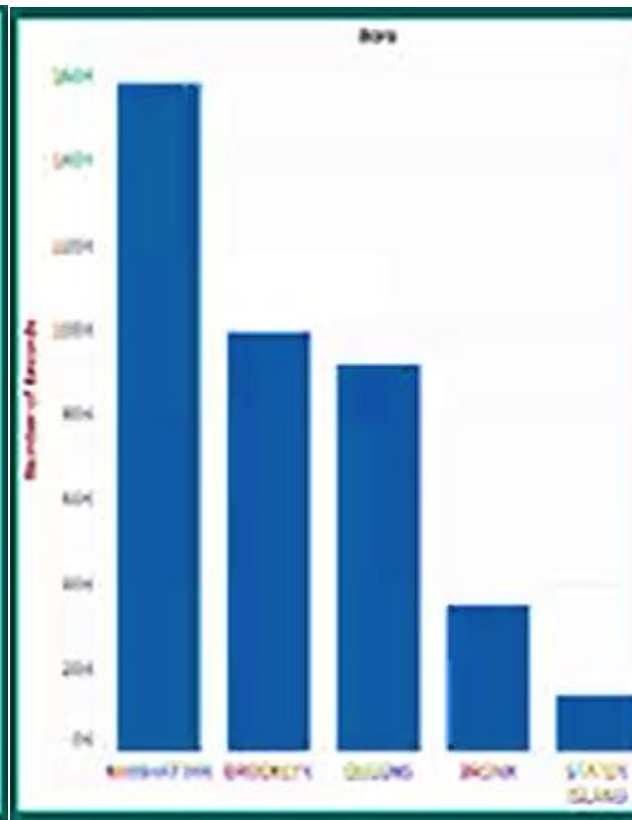
- Constraint view: mark type constrains what else can be encoded
 - **points**: 0 constraints on size, can encode more attributes w/ size & shape
 - **lines**: 1 constraint on size (length), can still size code other way (width)
 - **interlocking areas**: 2 constraints on size (length/width), cannot size or shape code
 - interlocking: size, shape, position
- Quick check: can you size-code another attribute
 - or is size/shape in use?

- Simplifying assumptions: one mark per item, single view
- Later on
 - multiple views
 - multiple marks in a region (glyph)
 - some items not represented by marks (aggregation and filtering)

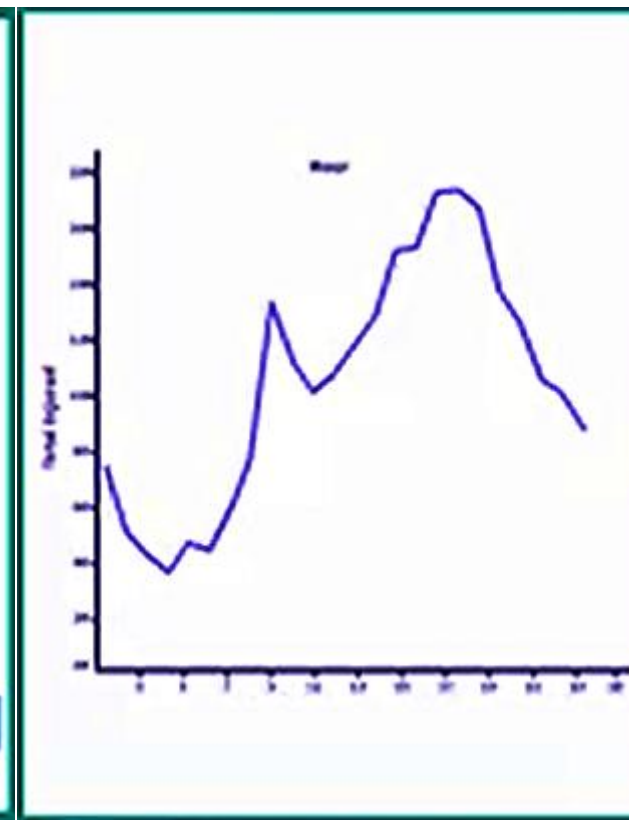
Marks: Four Common Types



POINTS



BARS



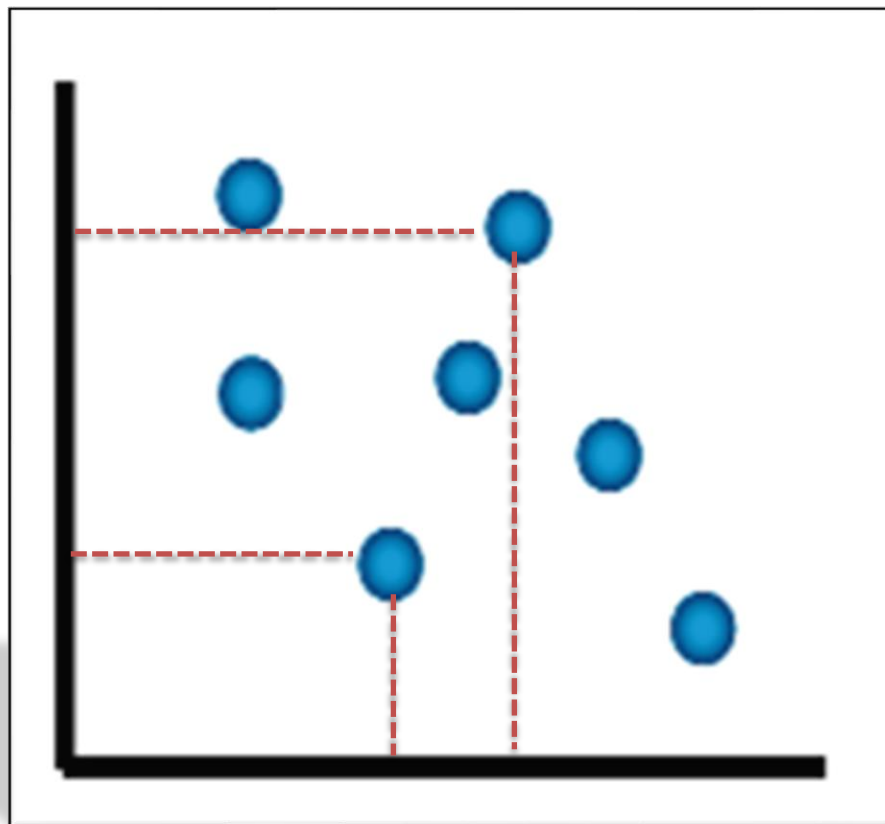
LINES



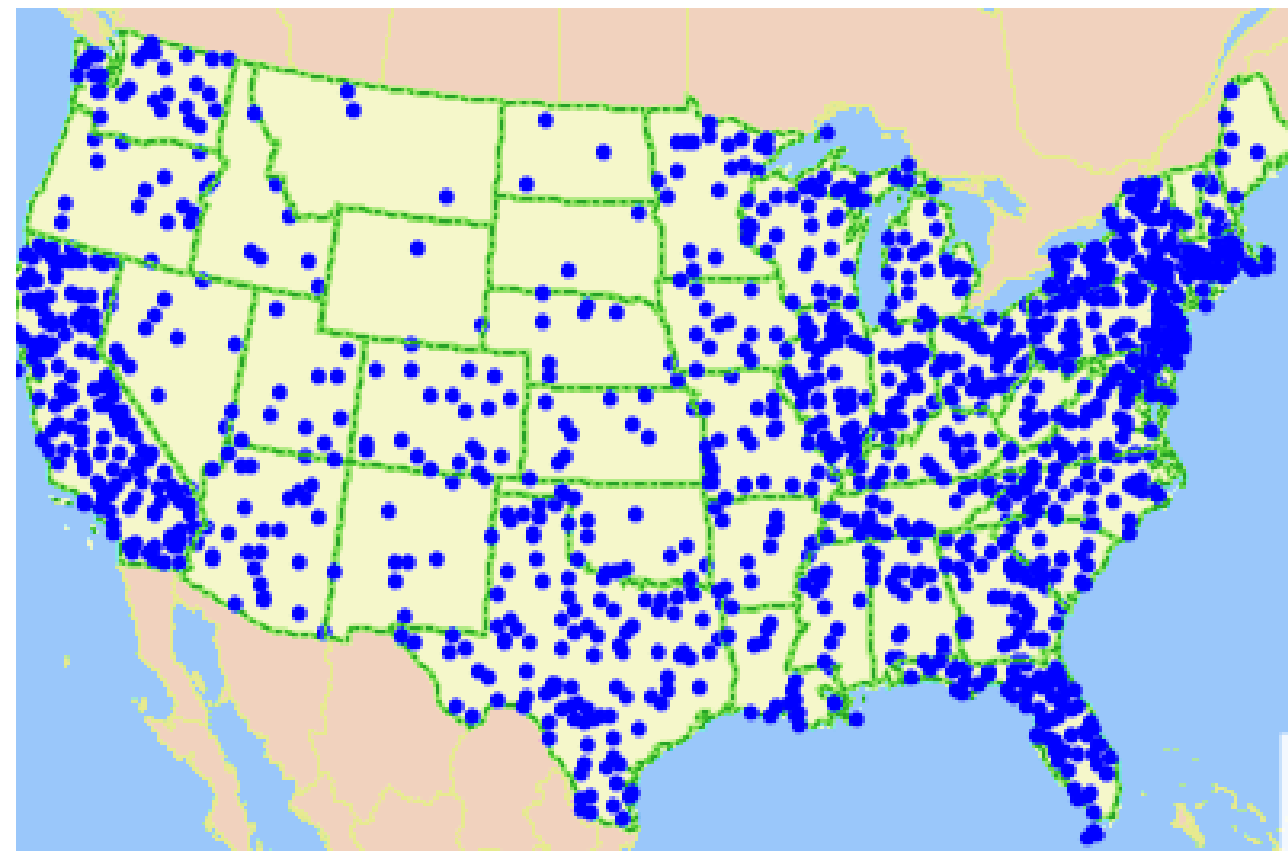
AREAS

Examples of Marks - Points

SCATTER PLOT

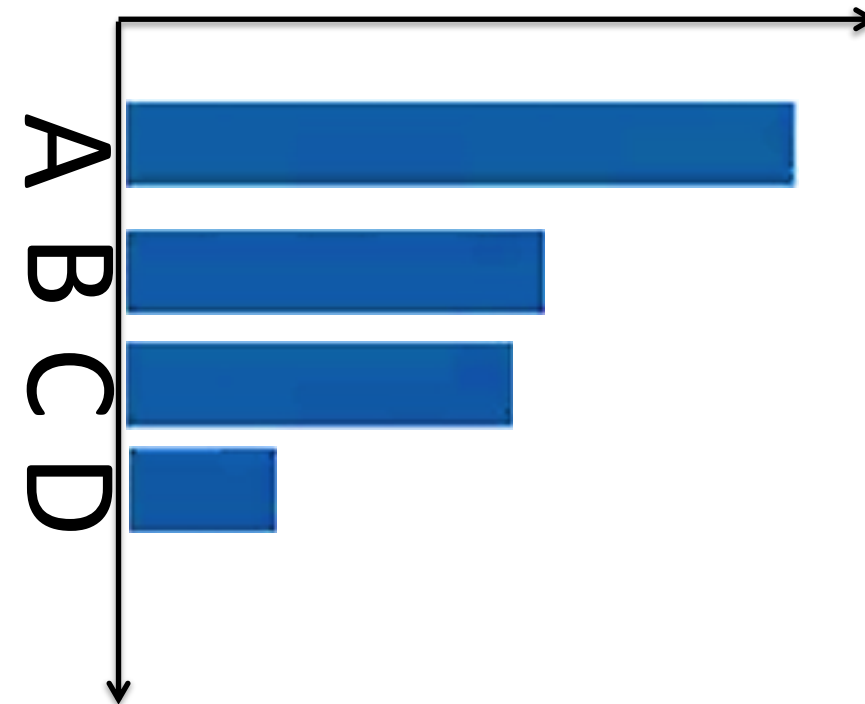
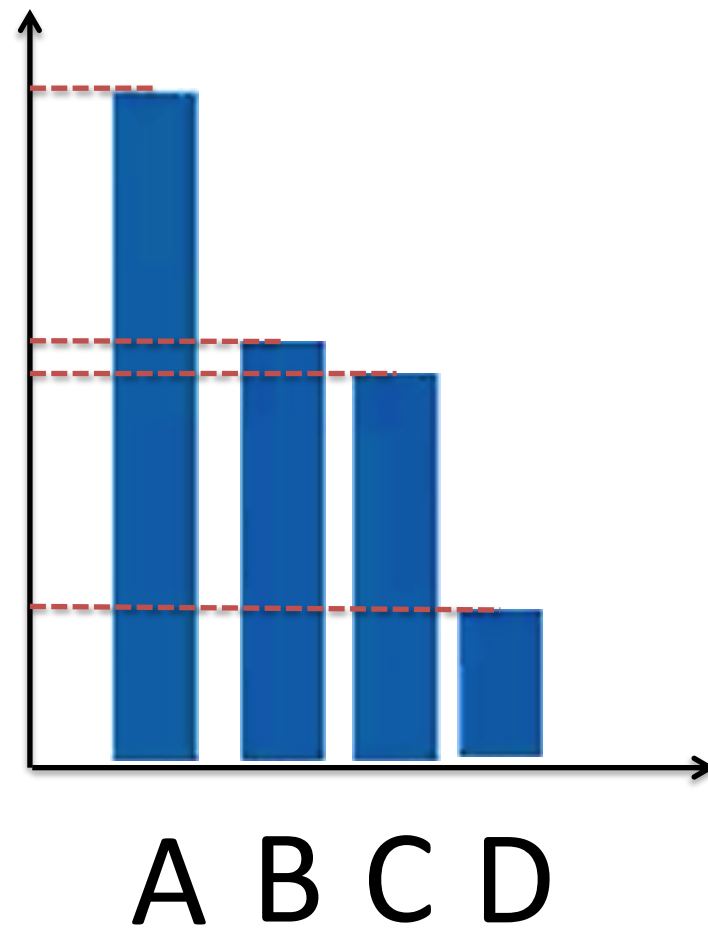


MAP



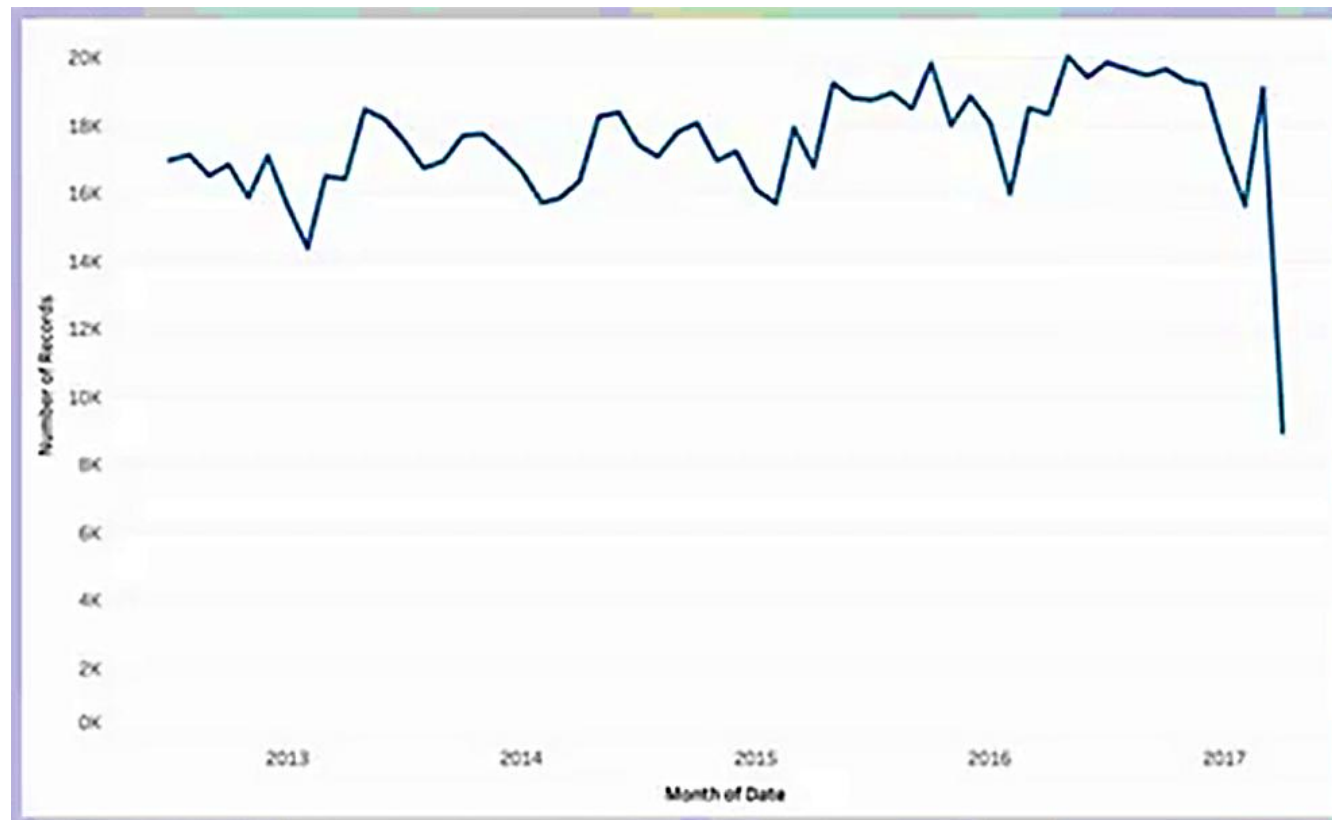
Examples of Marks - Bars

BARS CHART

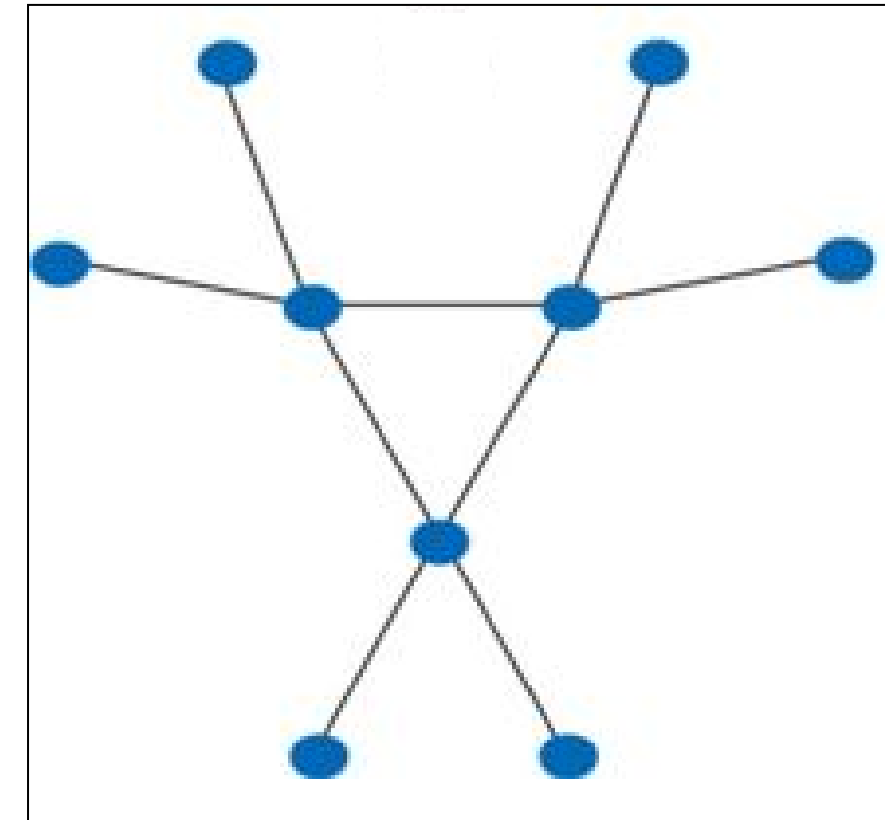


Examples of Marks - Lines

LINES CHART

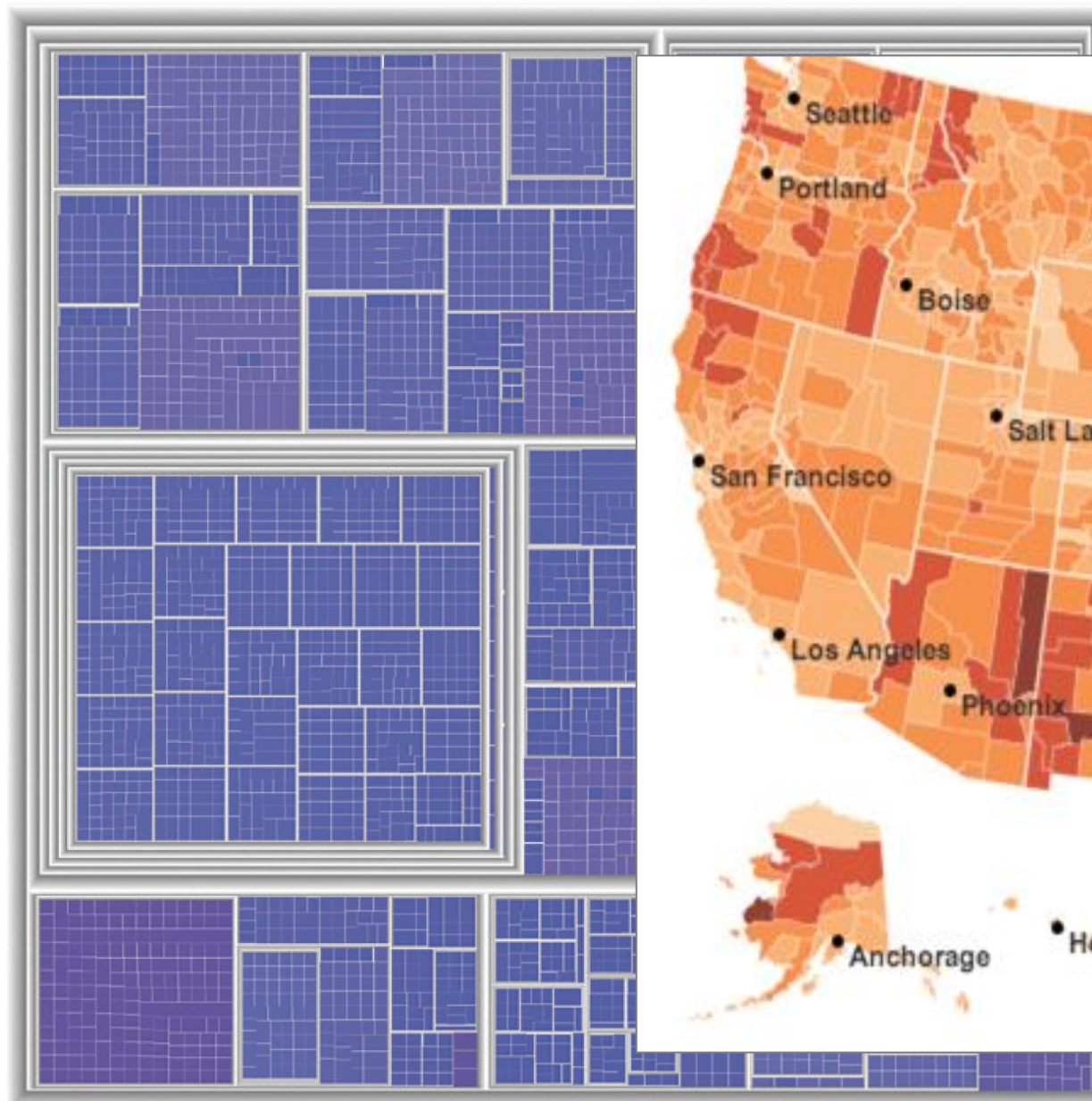


NETWORK

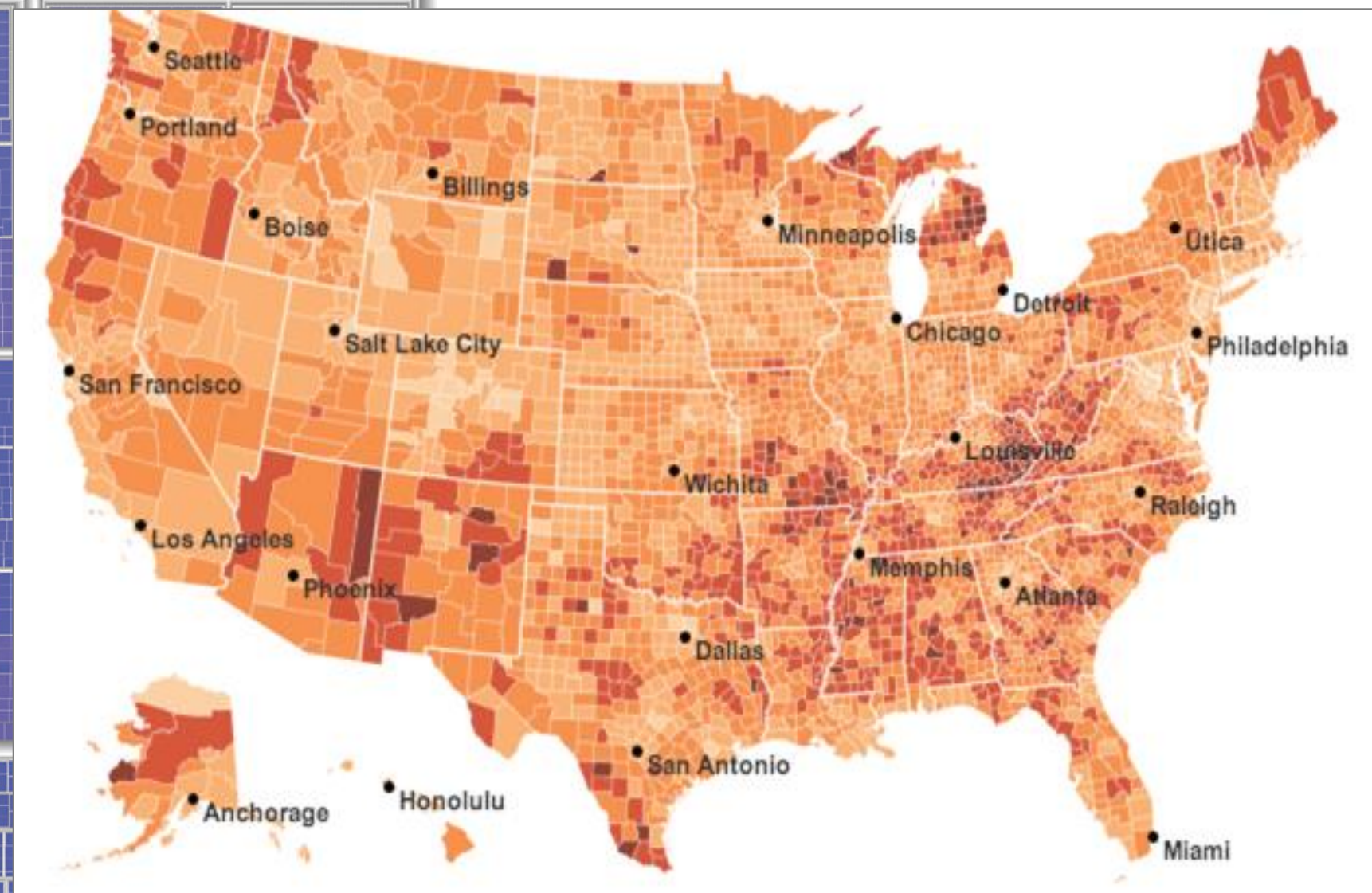


Examples of Marks - Areas

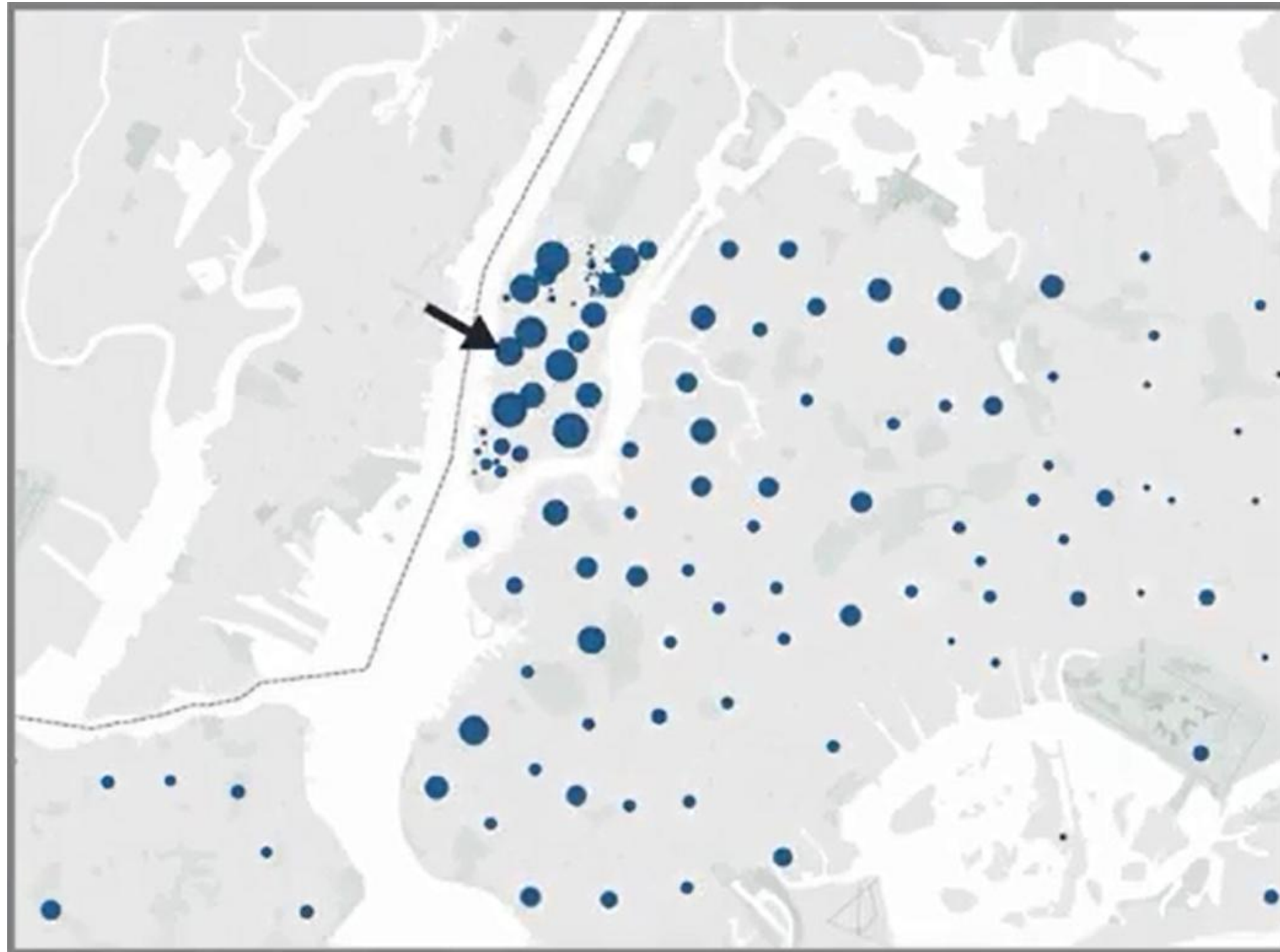
TREEMAP



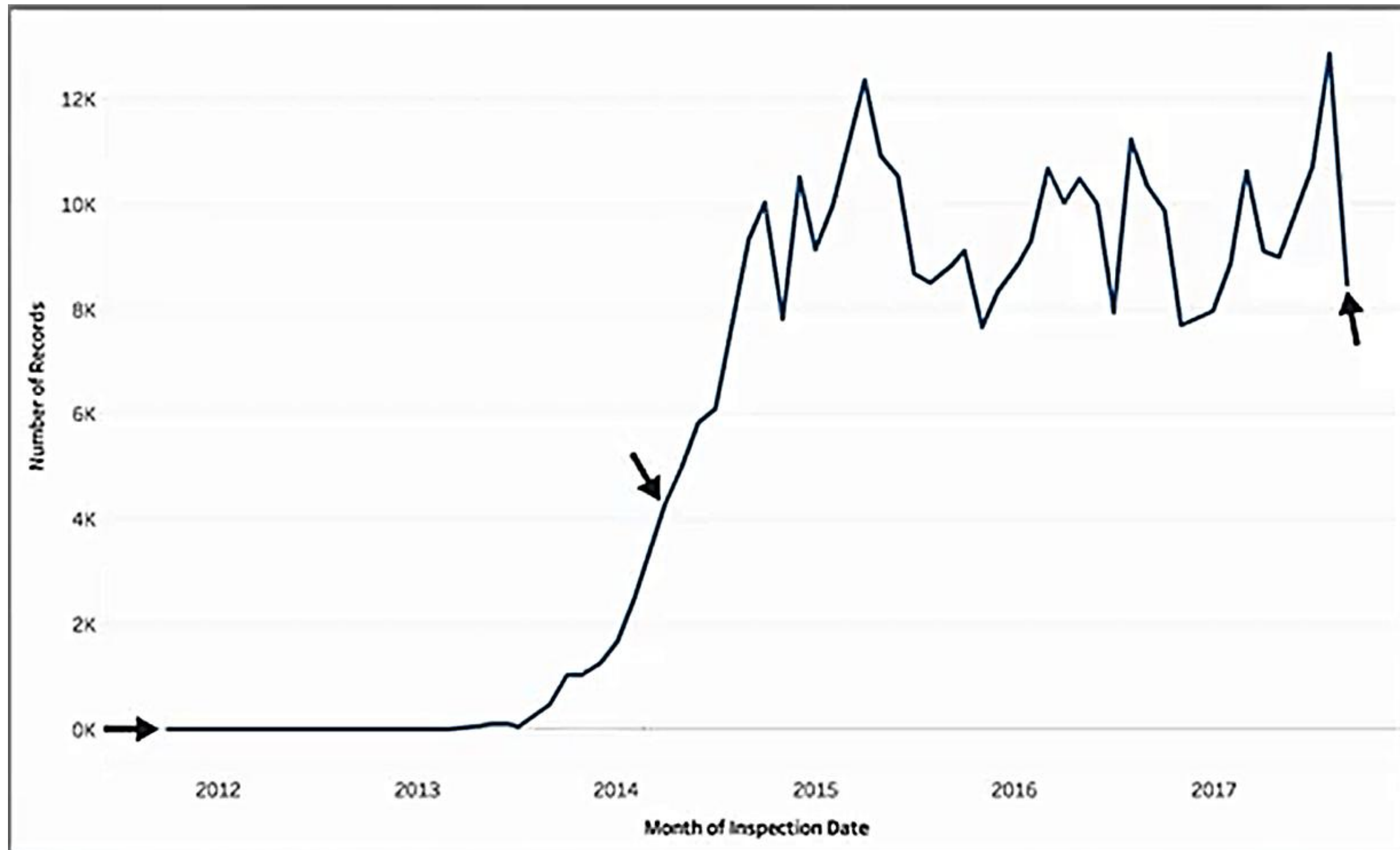
MAP



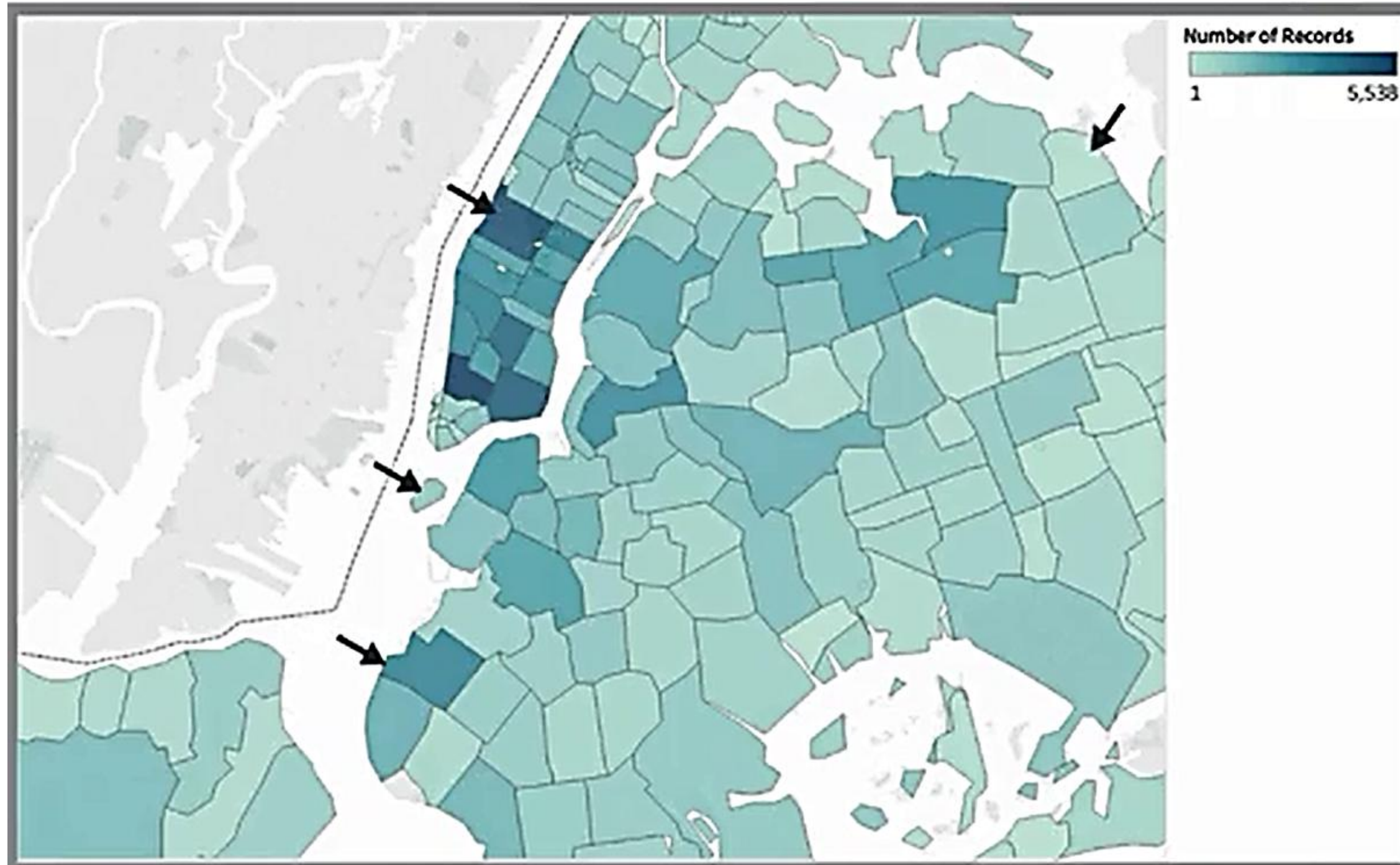
Visual Encodings – Simple Example



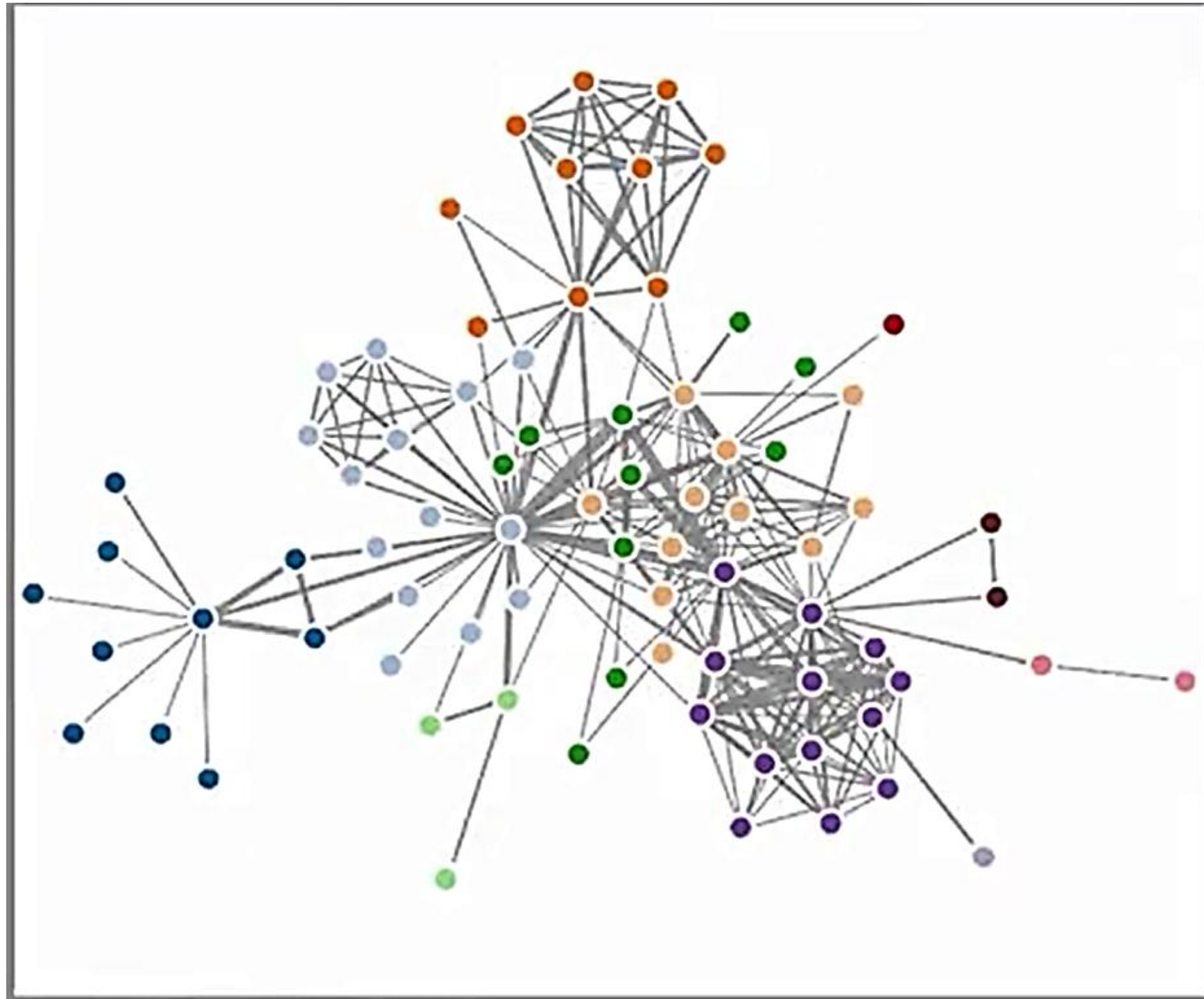
Visual Encodings – Simple Example



Visual Encodings – Real Example

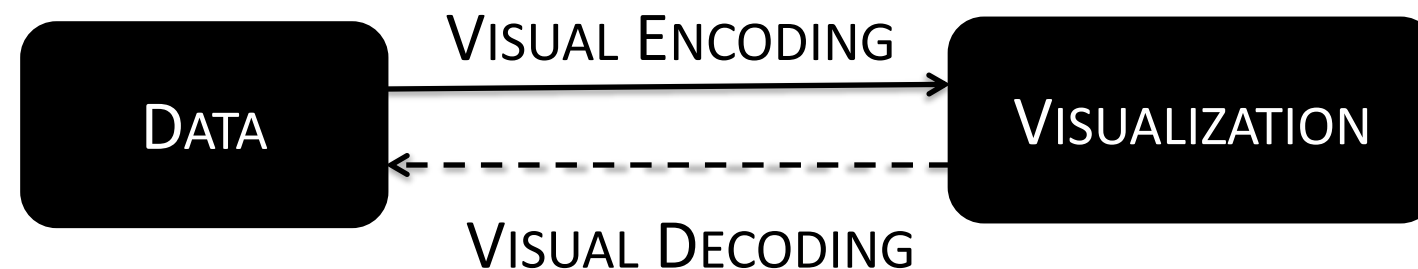


Visual Encodings – Real Example

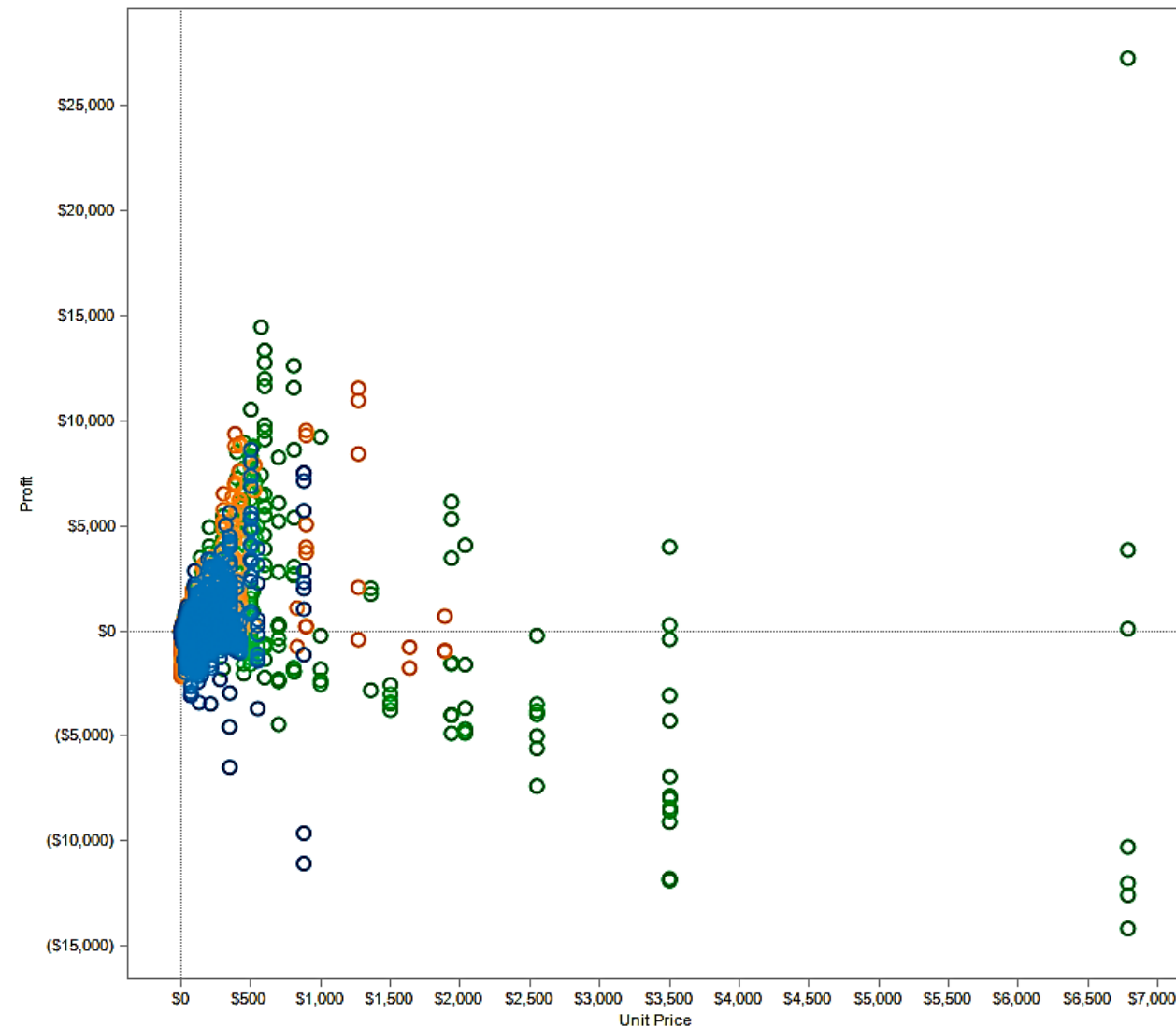


Graphical 'Decoding'

- The reverse of graphical encoding – looking at the vis idiom, try to reverse engineer its contents and map into their basic graphical contents e.g., marks and channels
- And then for each marks and channels identified, try to relate them to potential data items (objects) and its attributes.



Decoding Example



Data Items: Sales

Data Attributes: Price, Profit, Product Type

Visual Mark: Point

Visual Channel: x-Position, y-Position, Color

Encoding Rules:

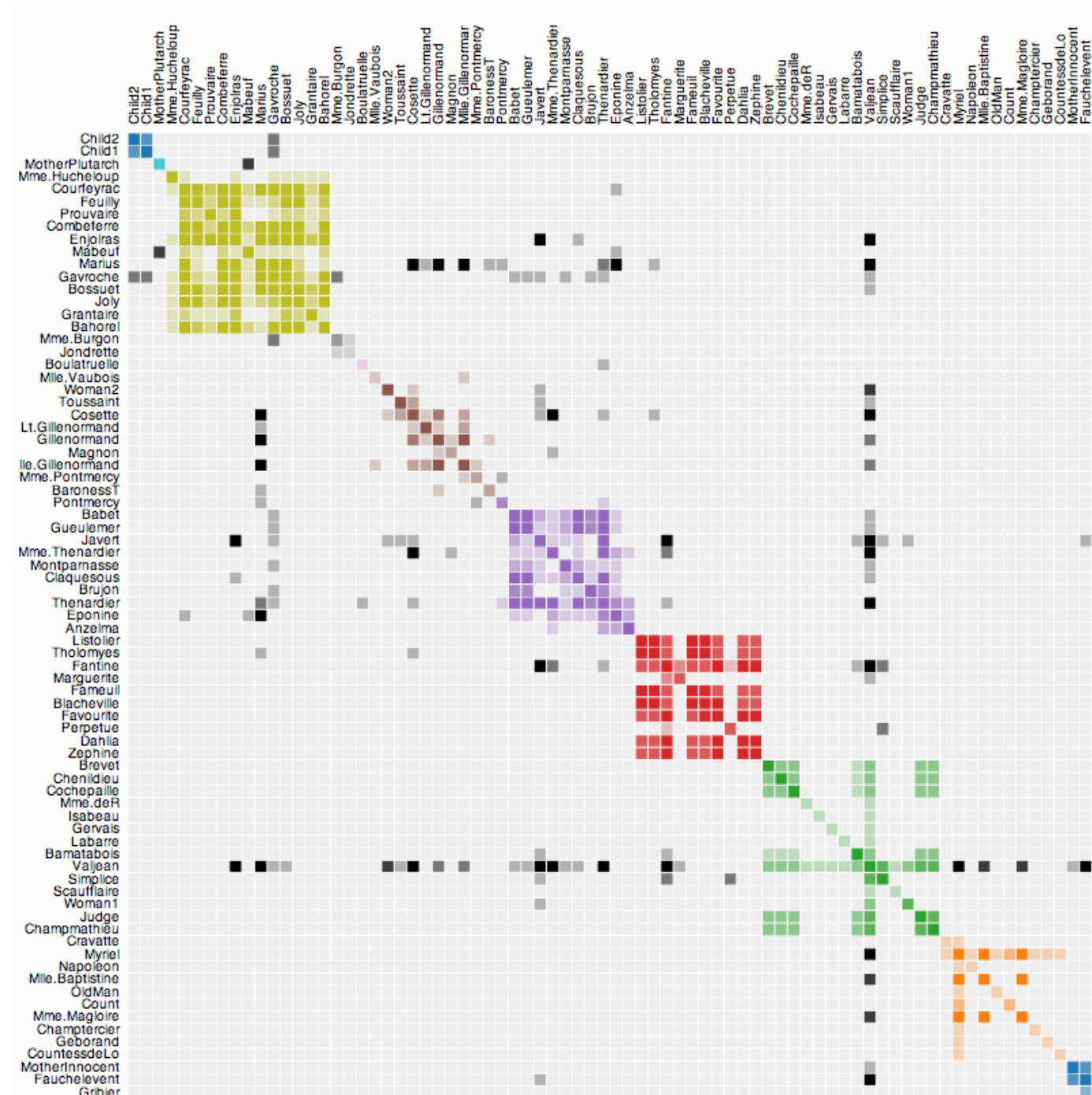
Sale = Point

Product Price and *Profit* = x,y-Positions

Product Type = Color

Decoding Example

Characters Co-occurrence in Les Misérables



Data Items: Co-occurrences

Visual Marks: Point

Attributes: Name 1, Name 2,
Clusters, Frequency

Position X, Position Y: Names
(categorical)

Color Hue: Clusters

Color Intensity: Frequency

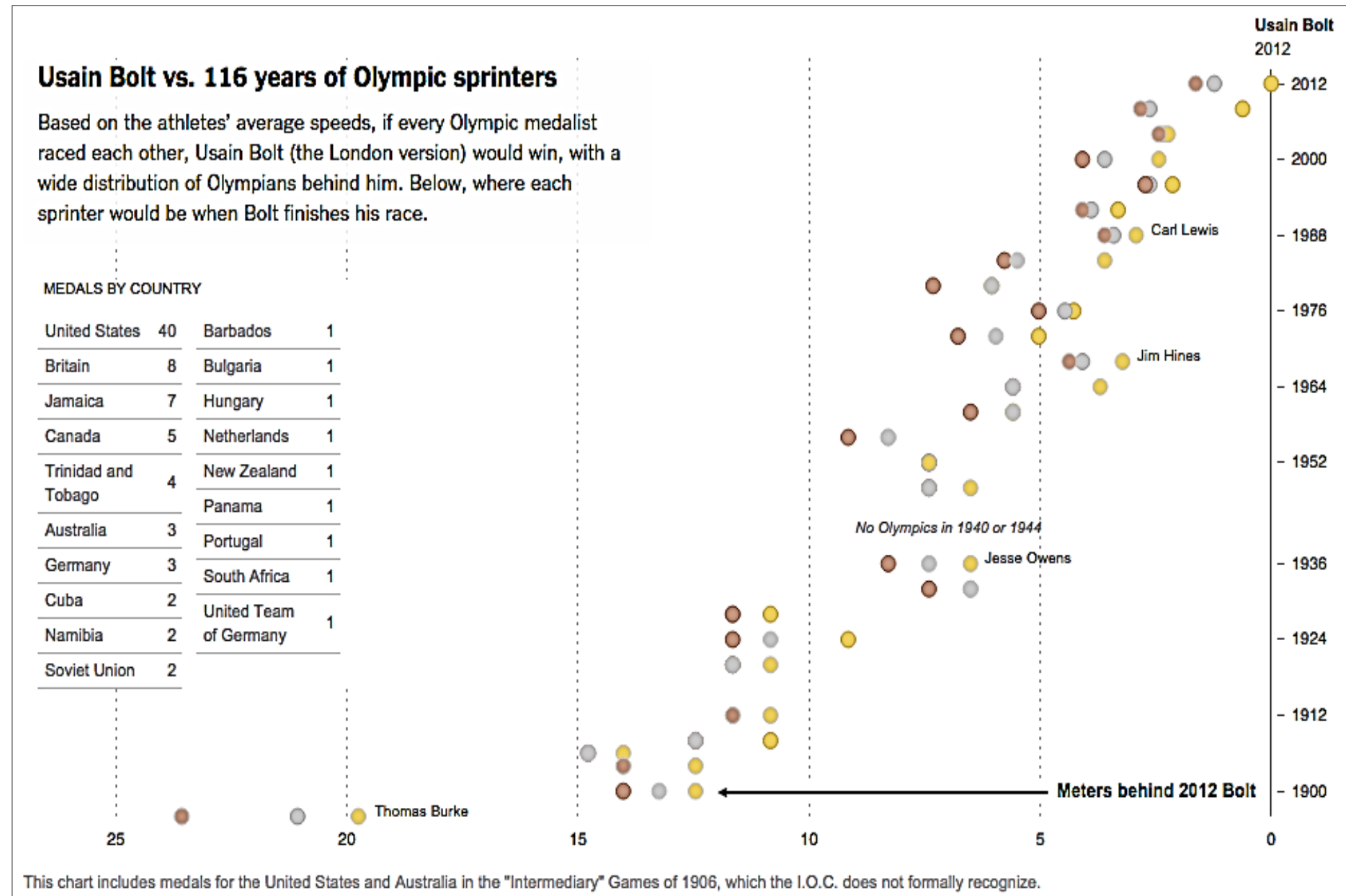
Class Activity (Marks & Channels)

Now it's your turn!

For each of the following identify:

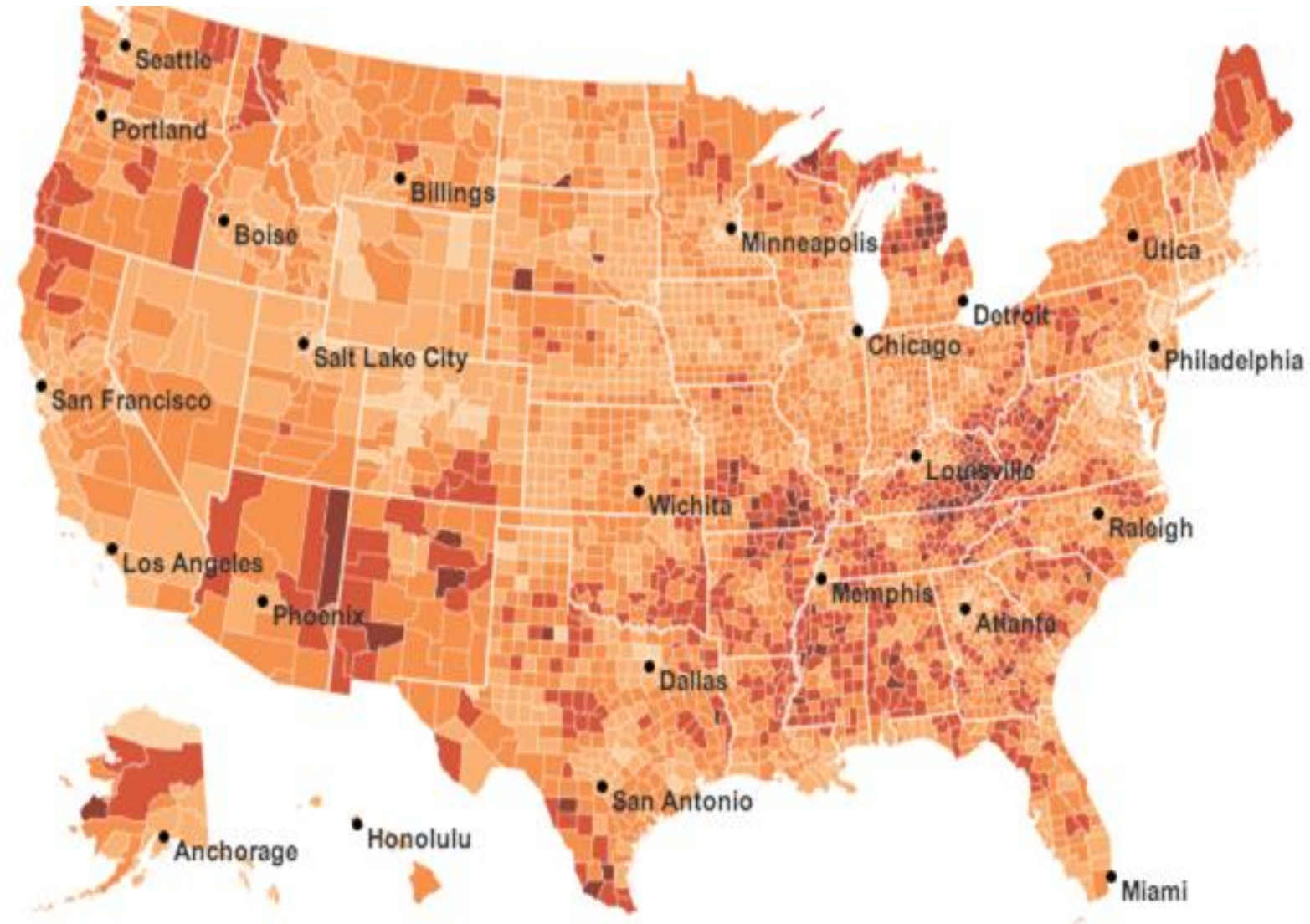
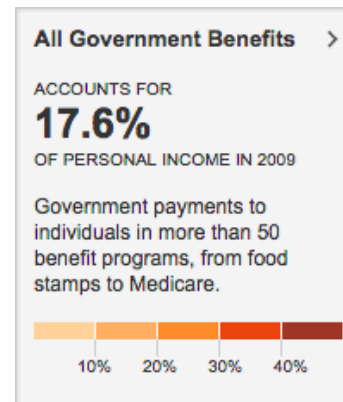
1. Data **Items** → Visual **Marks**
2. Data **Attributes** → Visual **Channels** Used

Class Activity (Marks & Channels): Visual 1



Source: http://www.nytimes.com/interactive/2012/08/05/sports/olympics/the-100-meter-dash-one-race-every-medalist-ever.html?_r=0

Class Activity (Marks & Channels): Visual 2



source: <http://www.nytimes.com/interactive/2012/02/12/us/entitlement-map.html>

How do we know how good the vis is?

- The main reason why we want to encode the data using particular graphical components is to achieve effective vis solution for the task.
- But how do we know whether the vis idiom is effective?
- There are TWO important principles to understand and reason whether the encoding is effective or not
 - **Expressiveness**
 - **Effectiveness**

Expressiveness

- match channel type to data type

Effectiveness

- some channels are better than others

Expressiveness Principle:

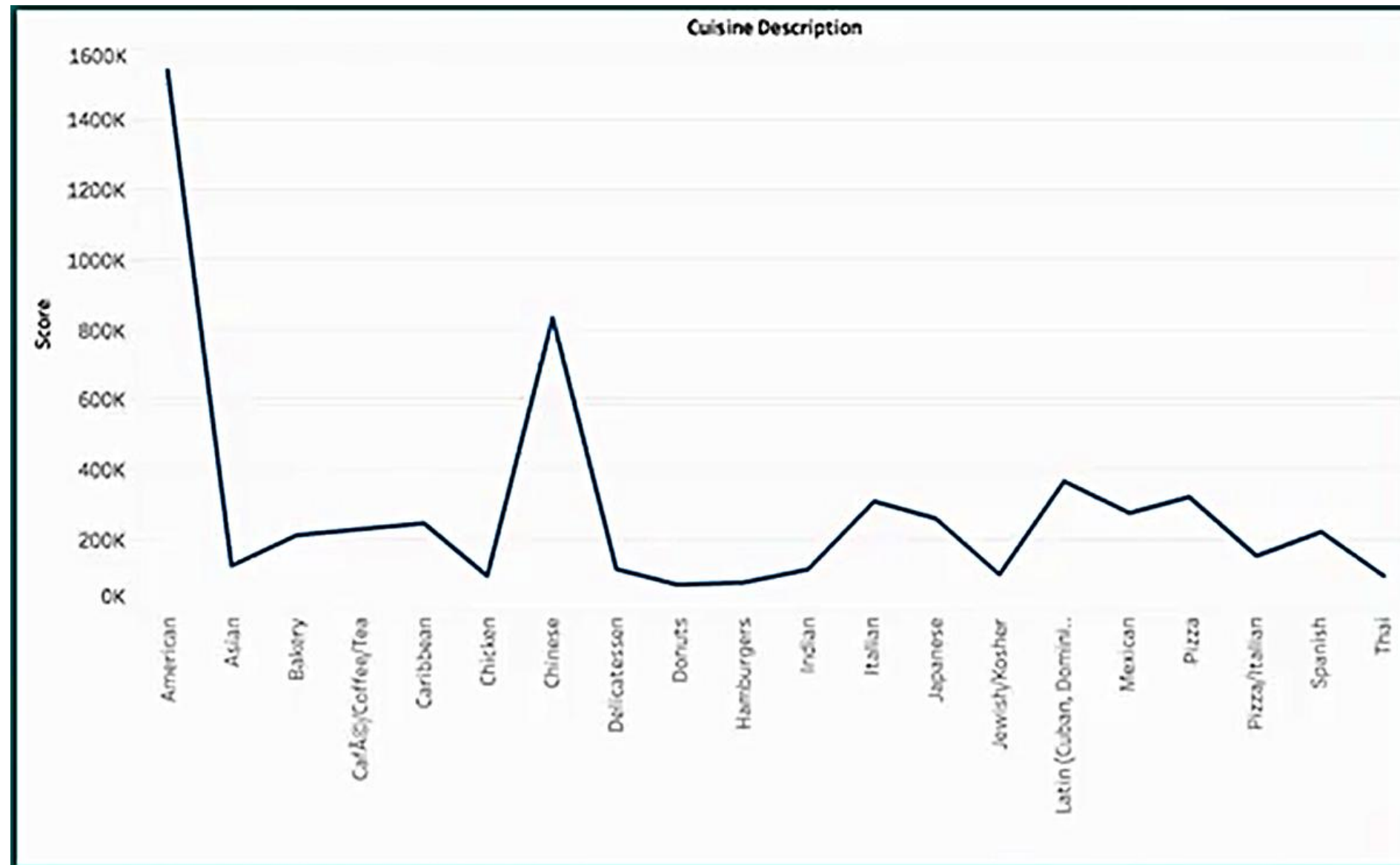
Visual information should express all and **only** the relationships or information that exist in the data.

e.g.,

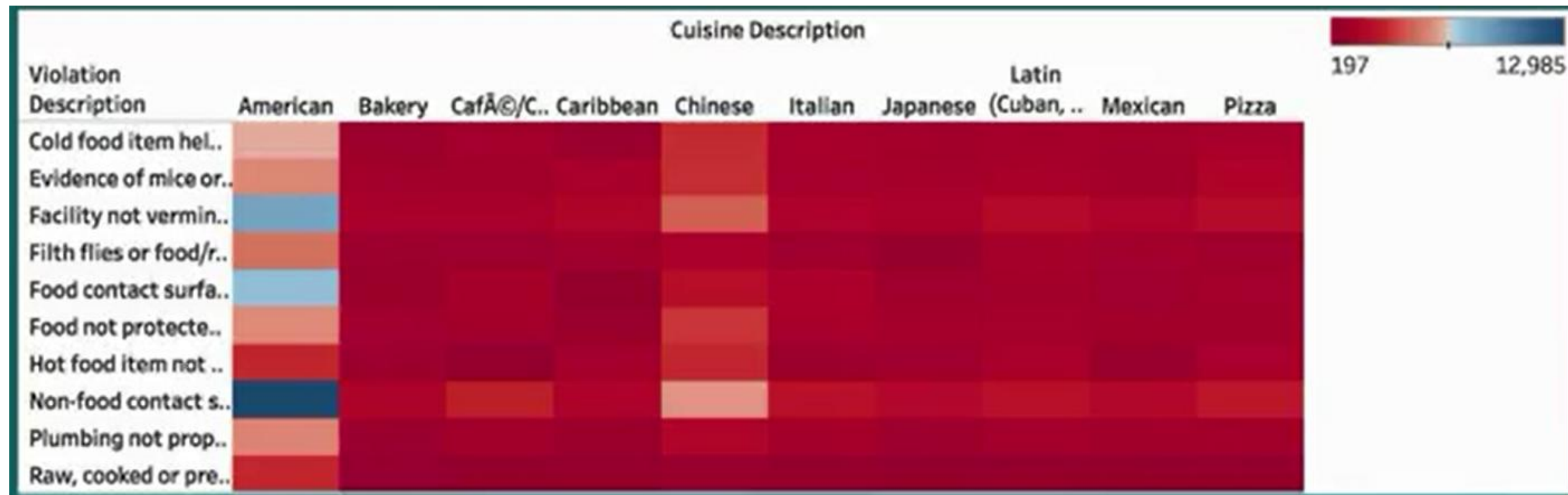
Ordered data should not appear as unordered.

Unordered data should not appear as ordered.

Good or Bad Viz?



Good or Bad Viz?



Matric Vis representing two **categorical attributes** on row & column and a quantitative attribute in each cell encoded using continuous colour hue + saturation (diverging colourmap)?

This gives the impression as if there are “middle” or average value or even some –ve values (the data have no such values)

The **effectiveness principle** dictates that the *importance* of the attribute should match the *salience* of the channel i.e., noticeability

... the most important attributes should be encoded with the most **effective** channels in order to be most noticeable.

What these examples illustrate is
different **visual channels** convey
Quantitative Magnitude differently
i.e., their effectiveness varies.

Two grouping of visual channels

Identity Channels:

information about *what, who, where* something is.

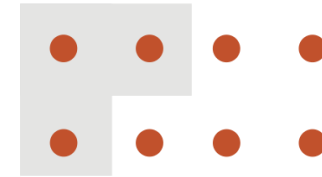
Magnitude Channels:

Information about *how much*.

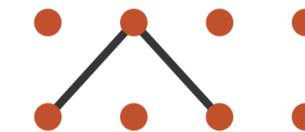
- Containment
- Connection
- Proximity
 - same spatial region
- Similarity
 - same values as other categorical channels

Marks as Links

➔ Containment



➔ Connection



➔ Identity Channels: **Categorical** Attributes

Spatial region



Color hue



Motion



Shape

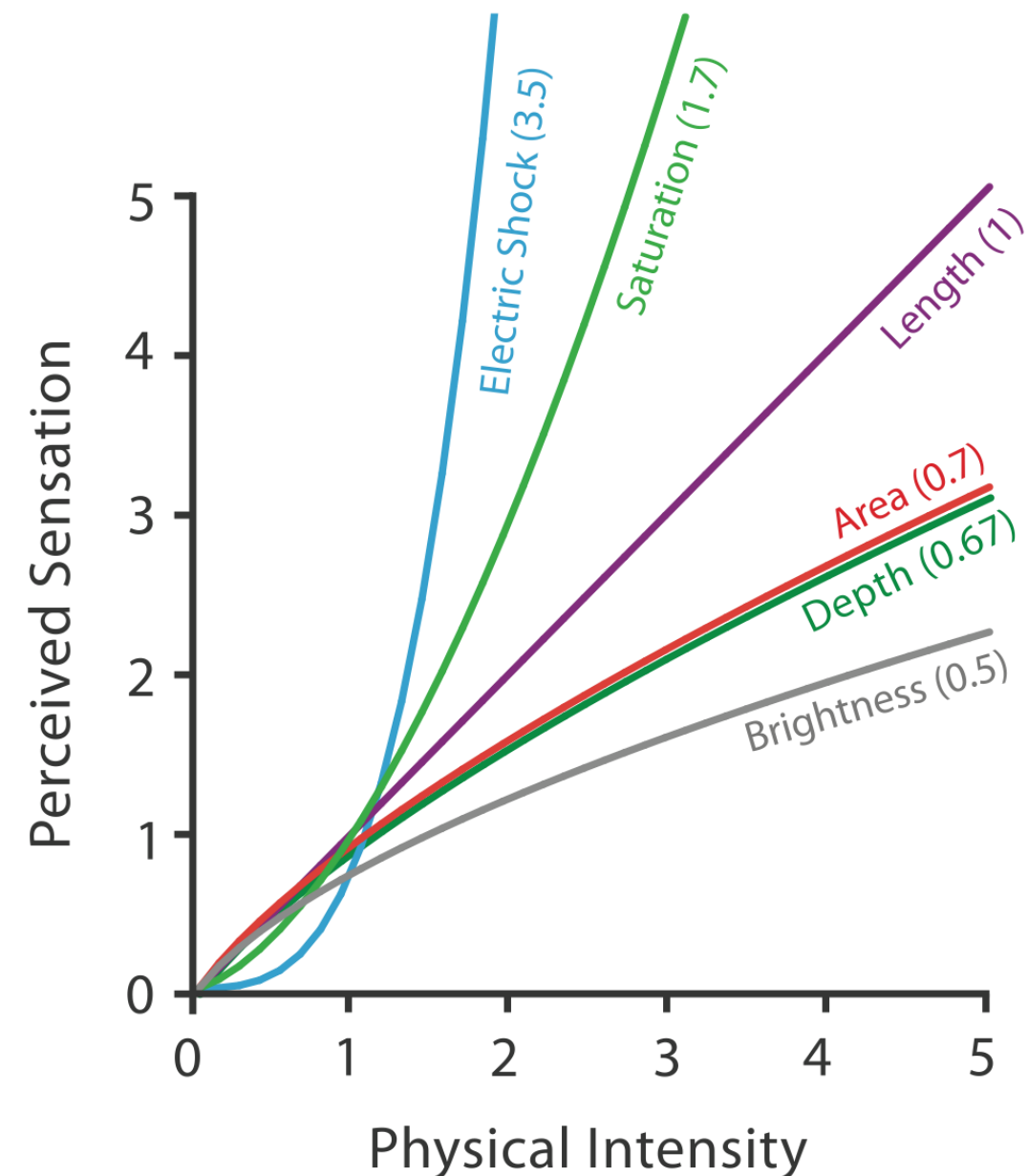


- **Accuracy:** how precisely can we tell the difference between encoded items?
- **Discriminability:** how many unique steps can we perceive?
- **Separability:** is our ability to use this channel affected by another one?
- **Popout:** can things jump out using this channel?

Accuracy: Fundamental theory

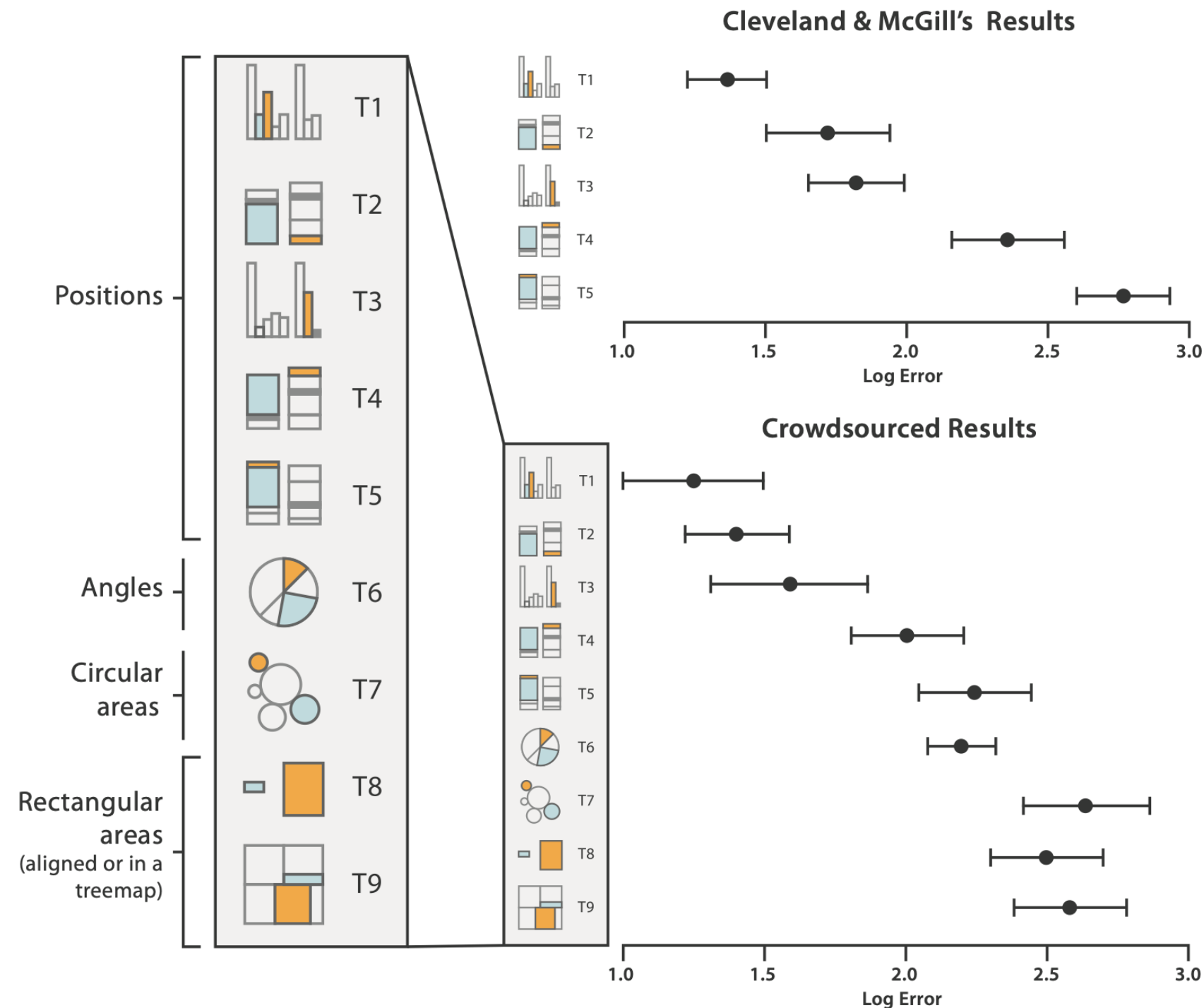
- Length is accurate: linear
- Others magnified or compressed
 - exponent characterizes

Steven's Psychophysical Power Law: $S = I^N$



S = sensation
 I = intensity

Accuracy: Vis experiments

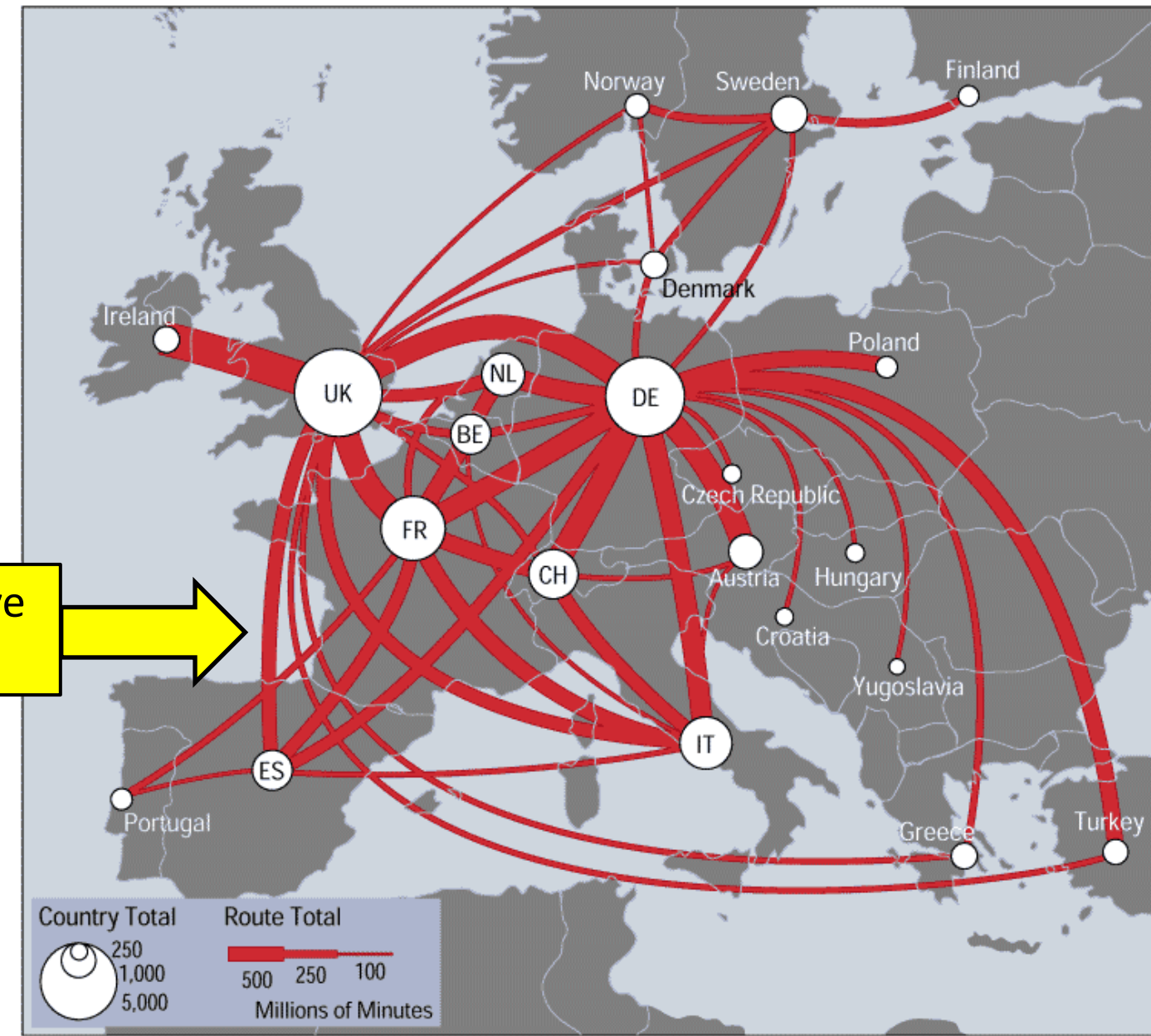


[Crowdsourcing Graphical Perception: Using Mechanical Turk to Assess Visualization Design. Heer and Bostock. Proc ACM Conf. Human Factors in Computing Systems (CHI) 2010, p. 203–212.]

Discriminability: How many usable values?

- Visual channel ability to support sufficient number of attribute levels/values to show
- Linewidth has only limited number of discriminable bins i.e., okay for smaller number of attribute's values

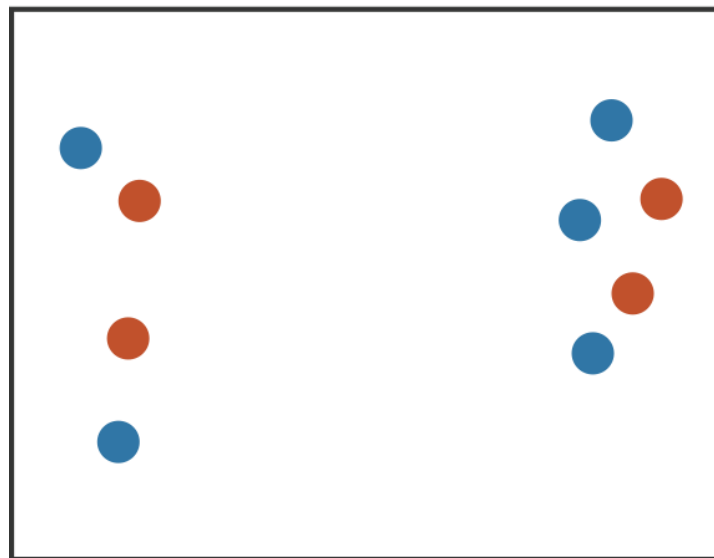
Example of effective linewidth



[\[mappa.mundi.net/maps/maps_014/telegeography.html\]](http://mappa.mundi.net/maps/maps_014/telegeography.html)

Separability vs. Integrality

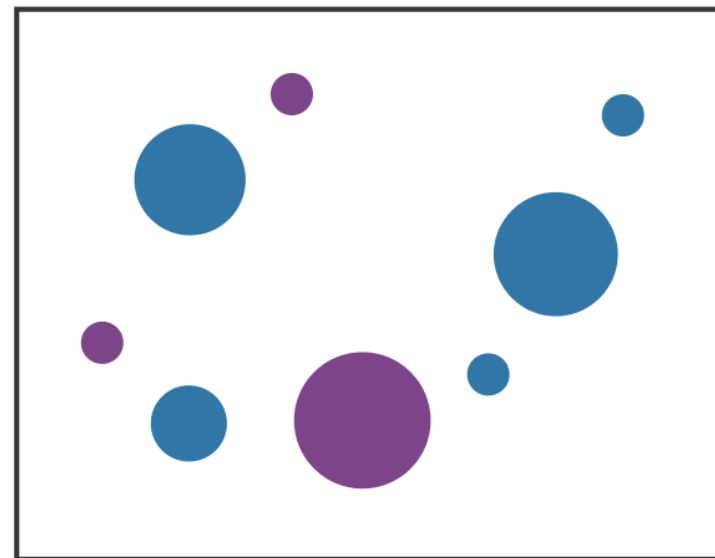
Position
+ Hue (Color)



Fully separable

2 groups each

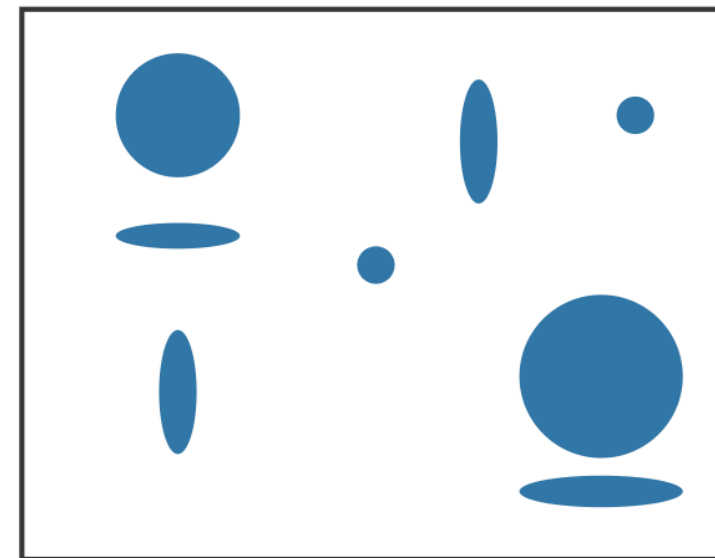
Size
+ Hue (Color)



Some interference

2 groups each

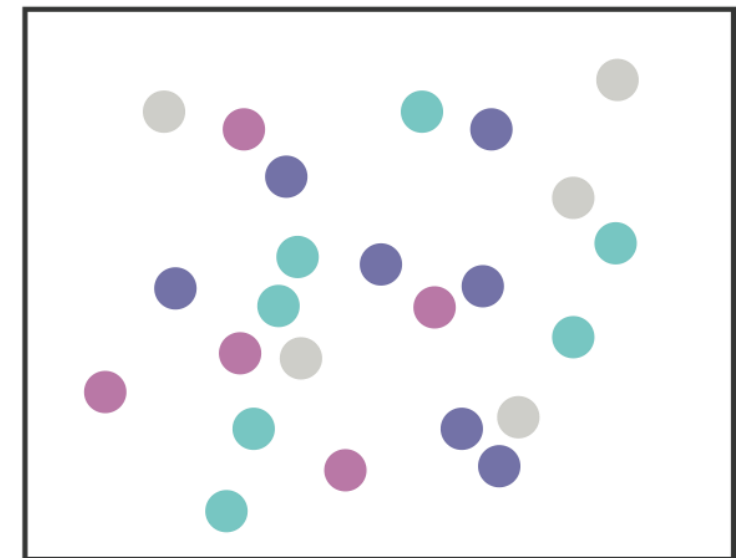
Width
+ Height



Some/significant
interference

3 groups total:
integral area

Red
+ Green

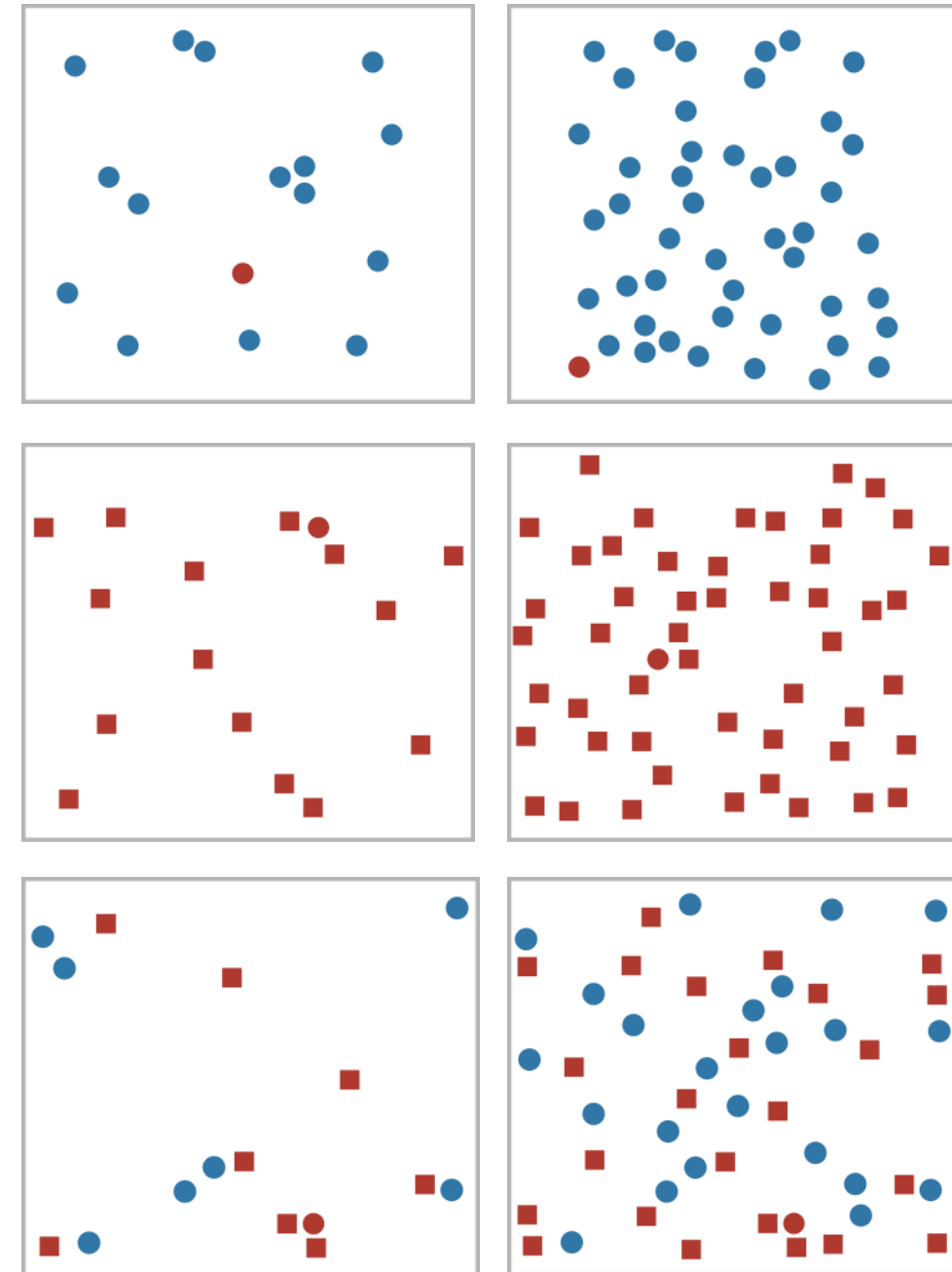


Major interference

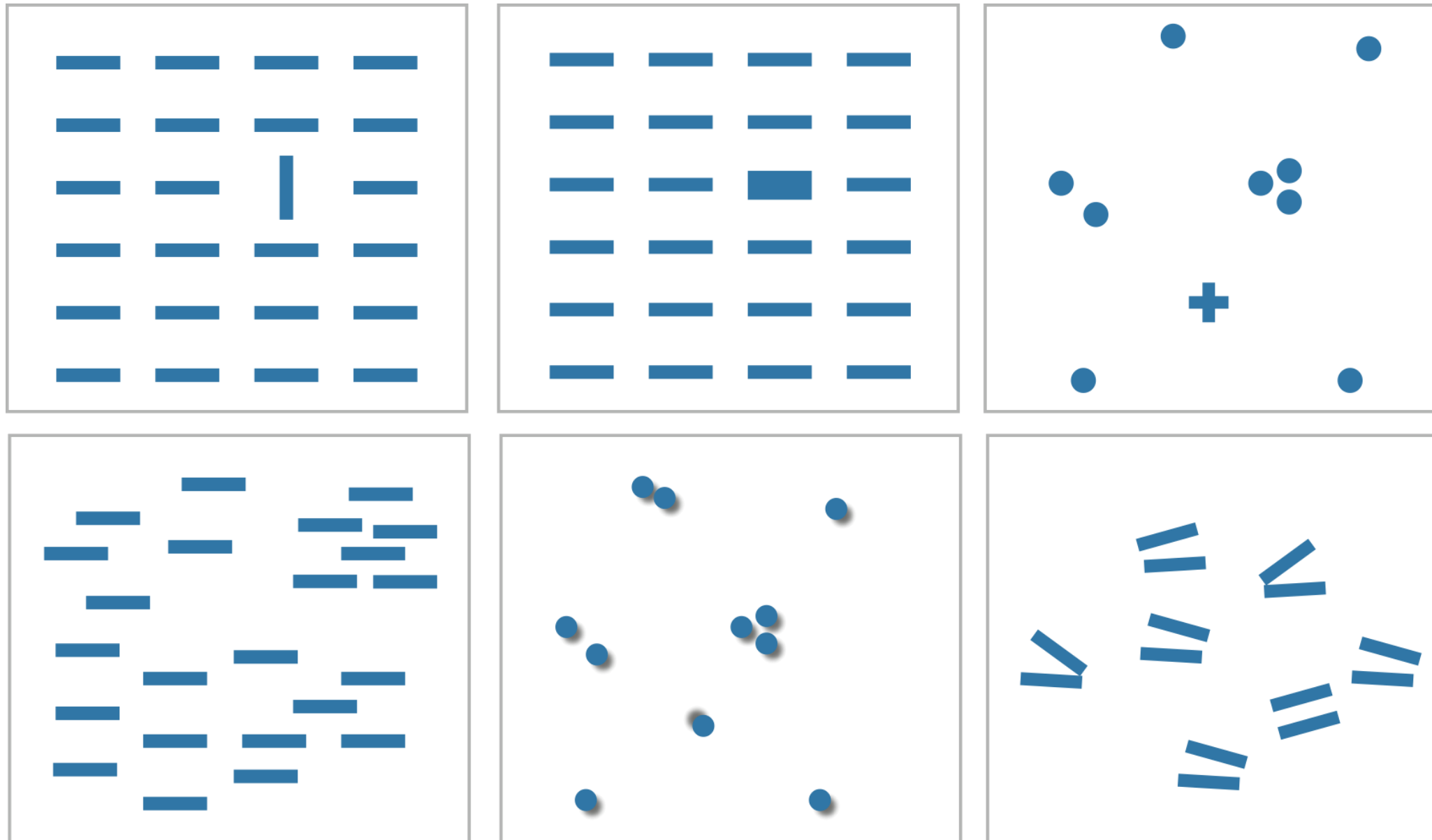
4 groups total:
integral hue

Popout

- Find the red dot
 - how long does it take?
- Parallel processing on many individual channels
 - speed independent of distractor count
 - speed depends on channel and amount of difference from distractors
- Serial search for (almost all) combinations
 - speed depends on number of distractors

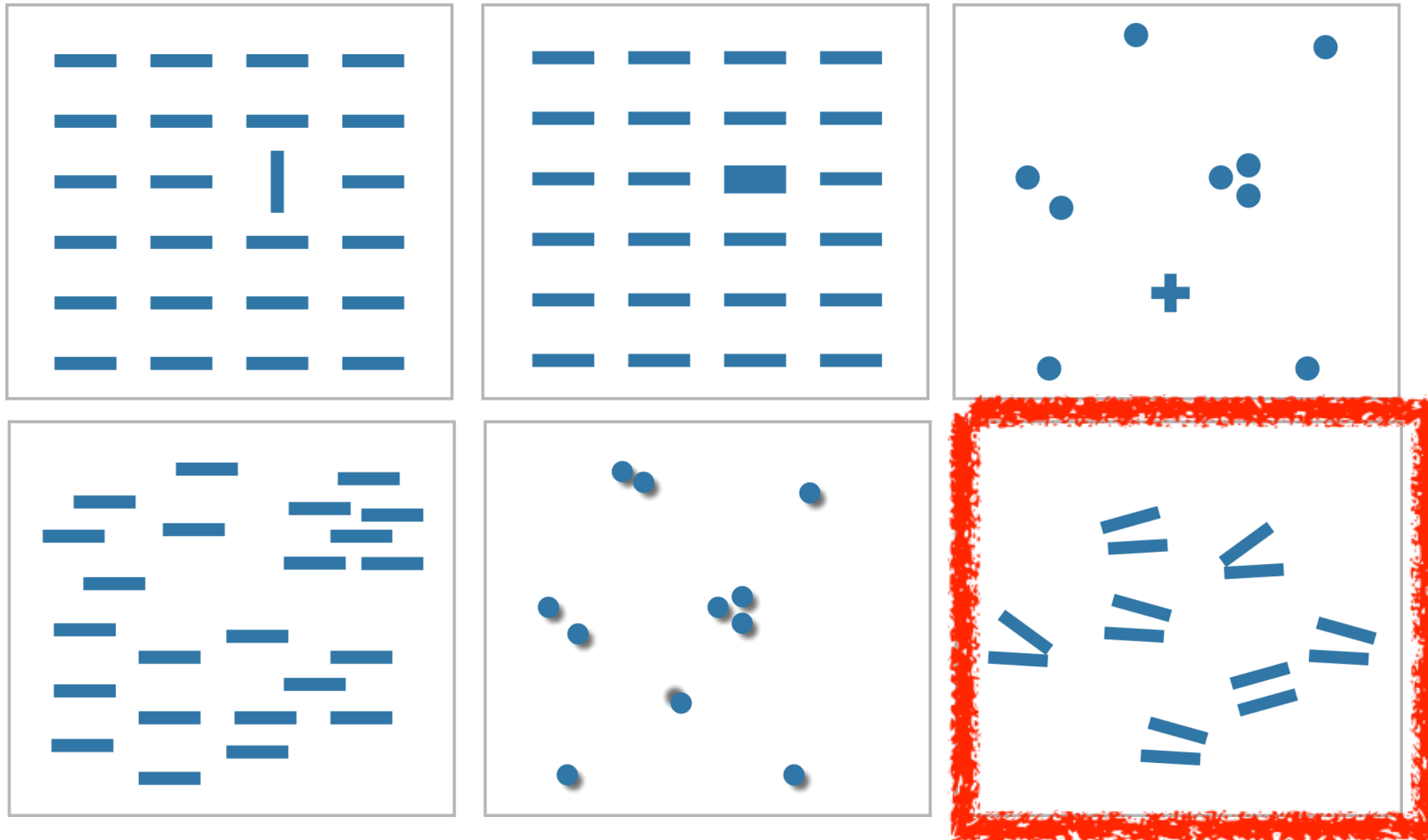


Popout



- Many channels
 - tilt, size, shape, proximity, shadow direction, ...

Popout



- Many channels
 - tilt, size, shape, proximity, shadow direction, ...
- but not all!
 - parallel line pairs do not pop out from tilted pairs

Factors affecting accuracy

- Alignment
- Distractors
- Distance
- Common scale / alignment

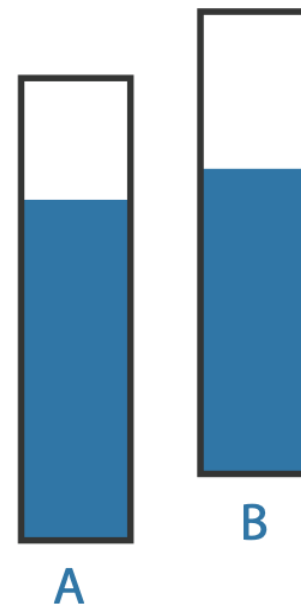


Relative vs. Absolute Judgements

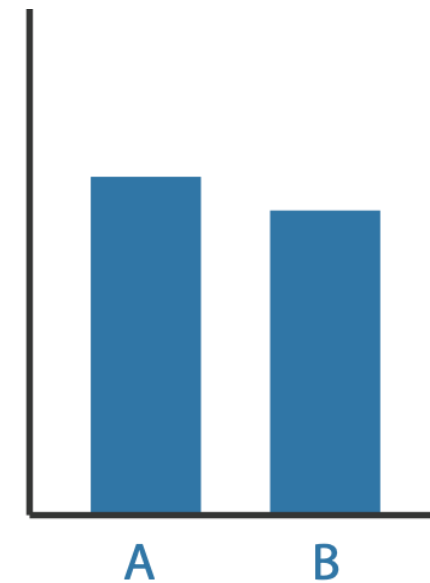
- Perceptual system mostly operates with relative judgements, not absolute
 - that's why accuracy increases with common frame/scale and alignment
 - Weber's Law: ratio of increment to background is constant
 - filled rectangles differ in length by 1:9, difficult judgement
 - white rectangles differ in length by 1:2, easy judgement



length



position along unaligned
common scale



position along aligned scale

Examples of effectiveness

For each of the upcoming examples, try to estimate *how much* 'bigger' is the value of B compared to the value of A

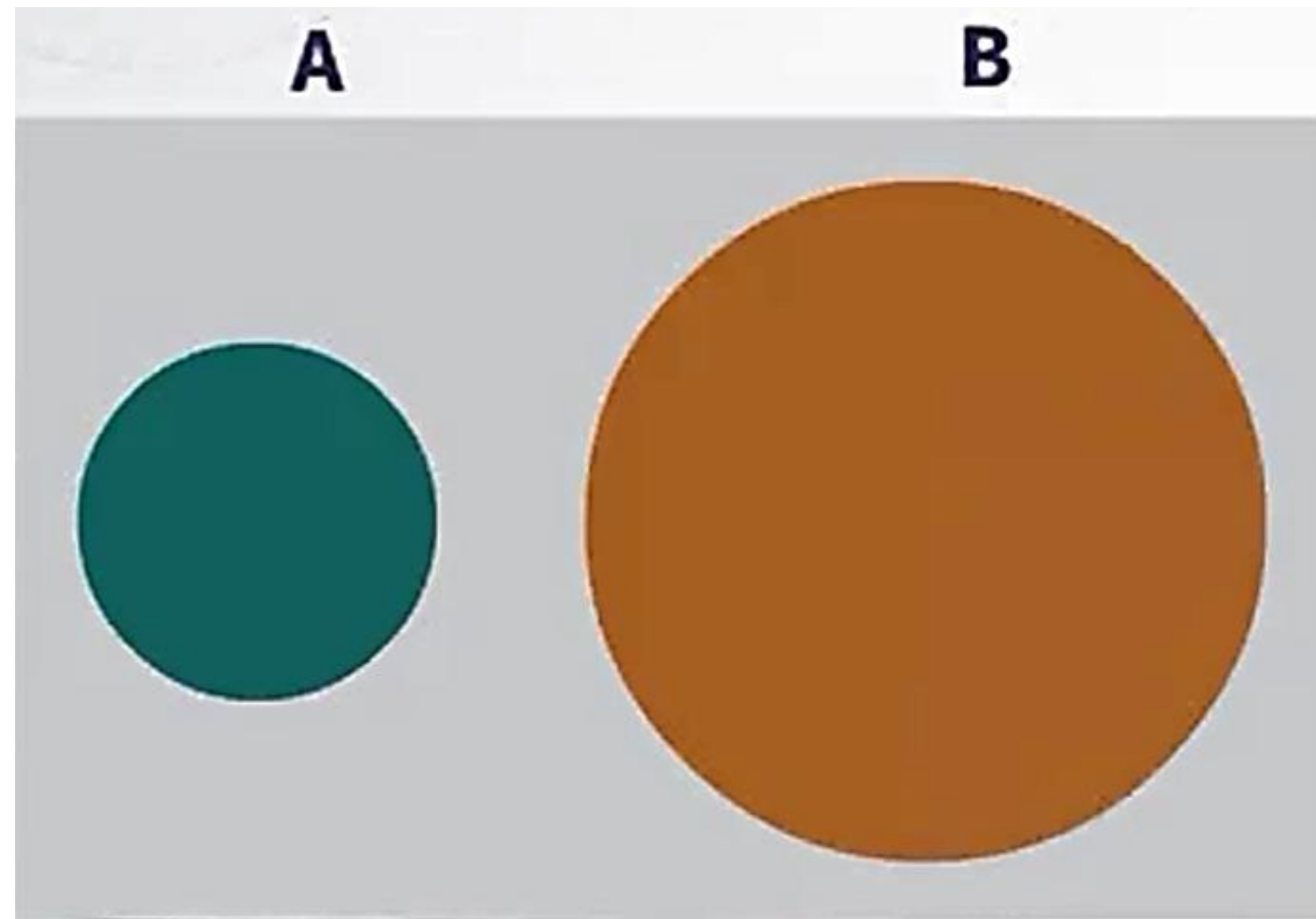
A

B



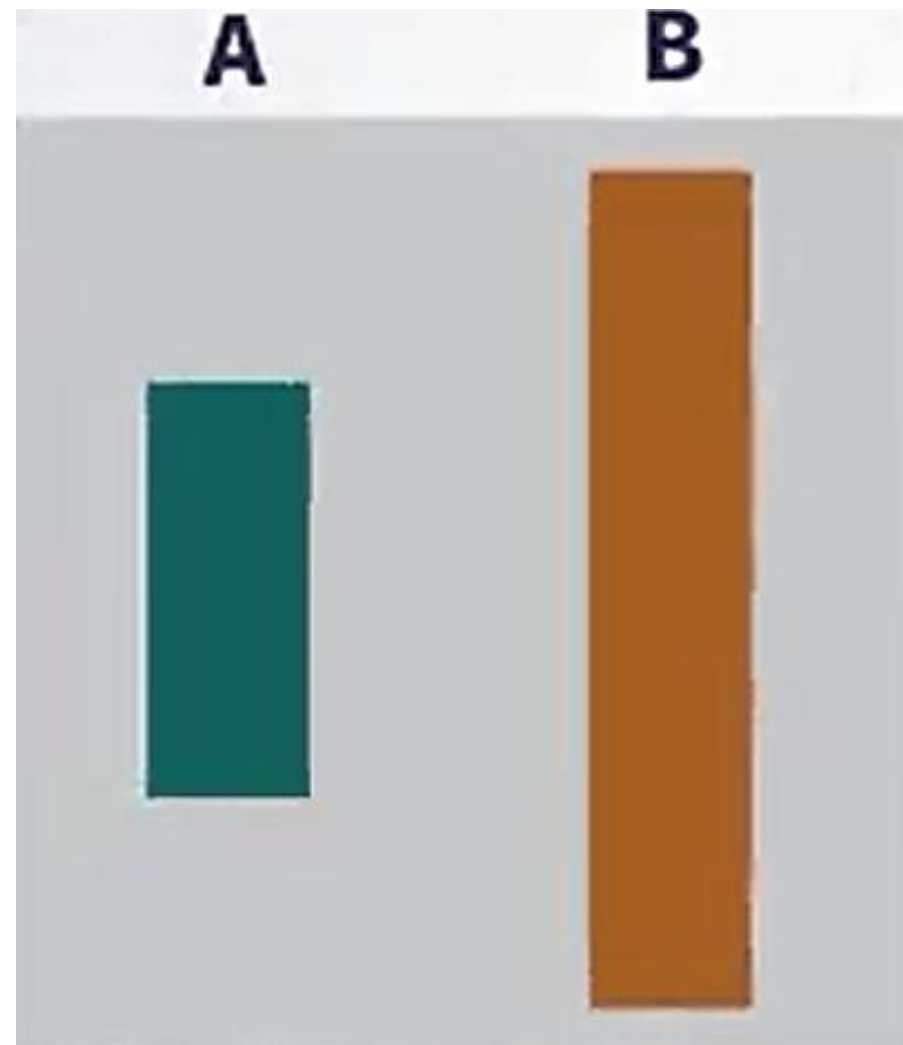
Examples of effectiveness

For each of the upcoming examples, try to estimate *how much* 'bigger' is the value of B compared to the value of A



Examples of effectiveness

For each of the upcoming examples, try to estimate *how much* 'bigger' is the value of B compared to the value of A



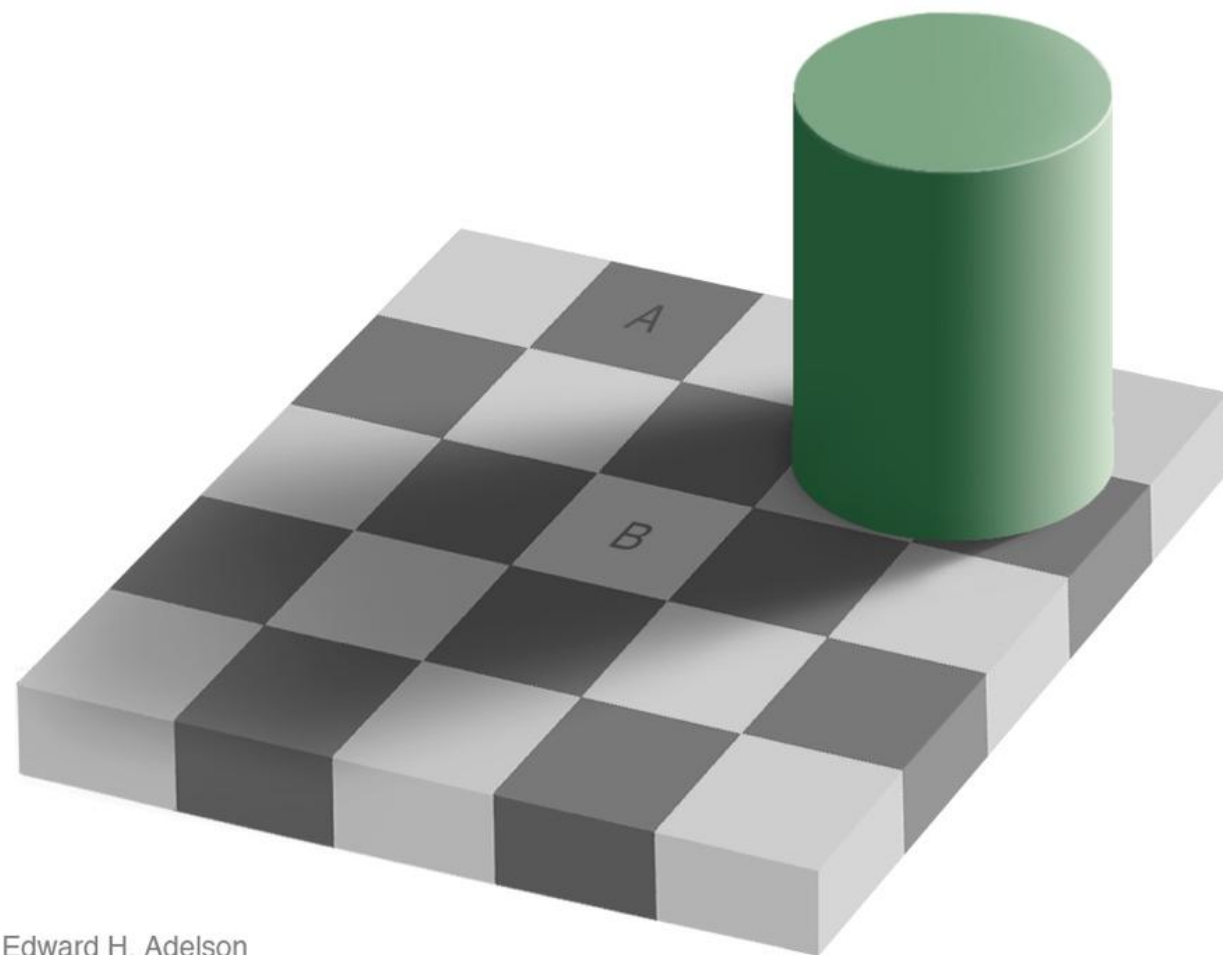
Examples of effectiveness

For each of the upcoming examples, try to estimate *how much* 'bigger' is the value of B compared to the value of A

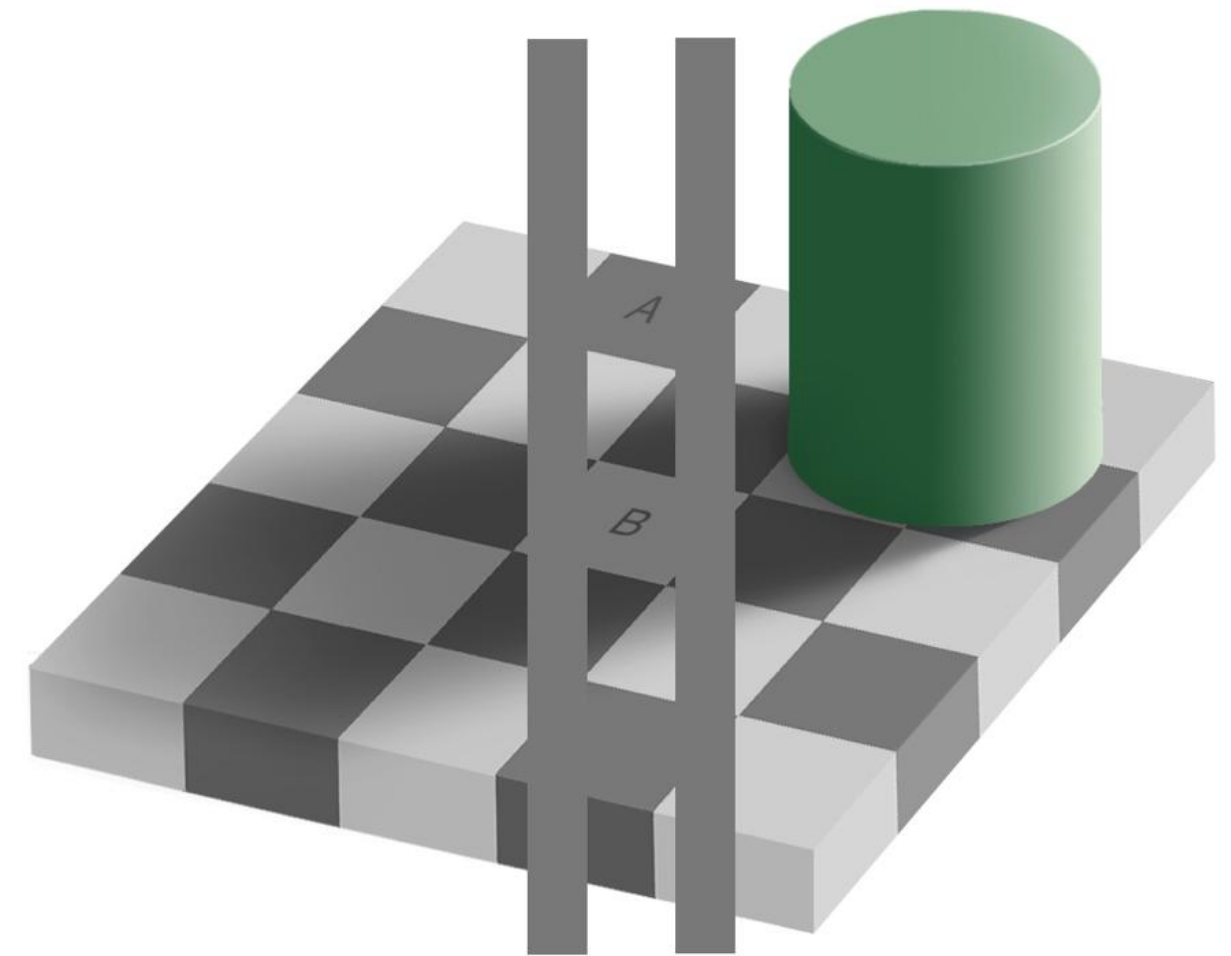


Relative luminance judgements

- Perception of luminance is contextual based on contrast with surroundings

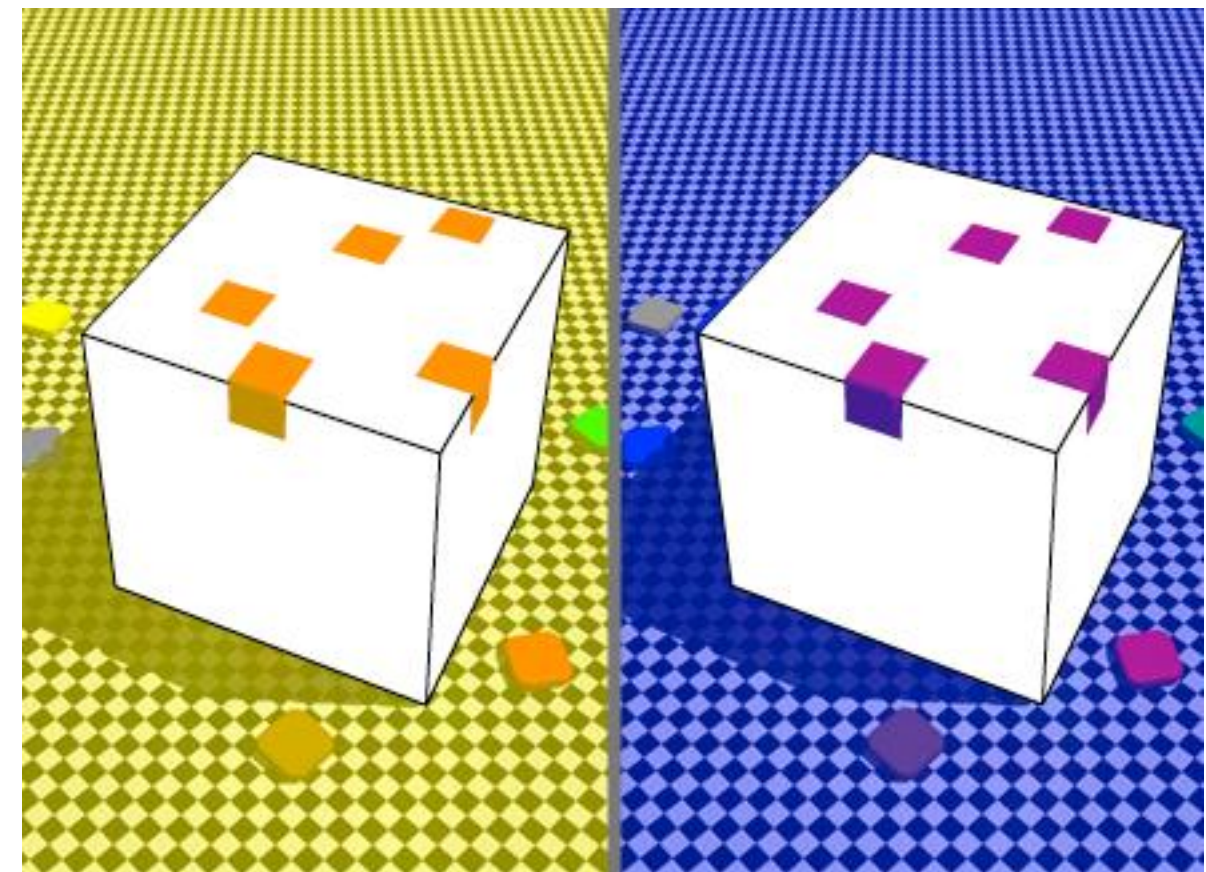
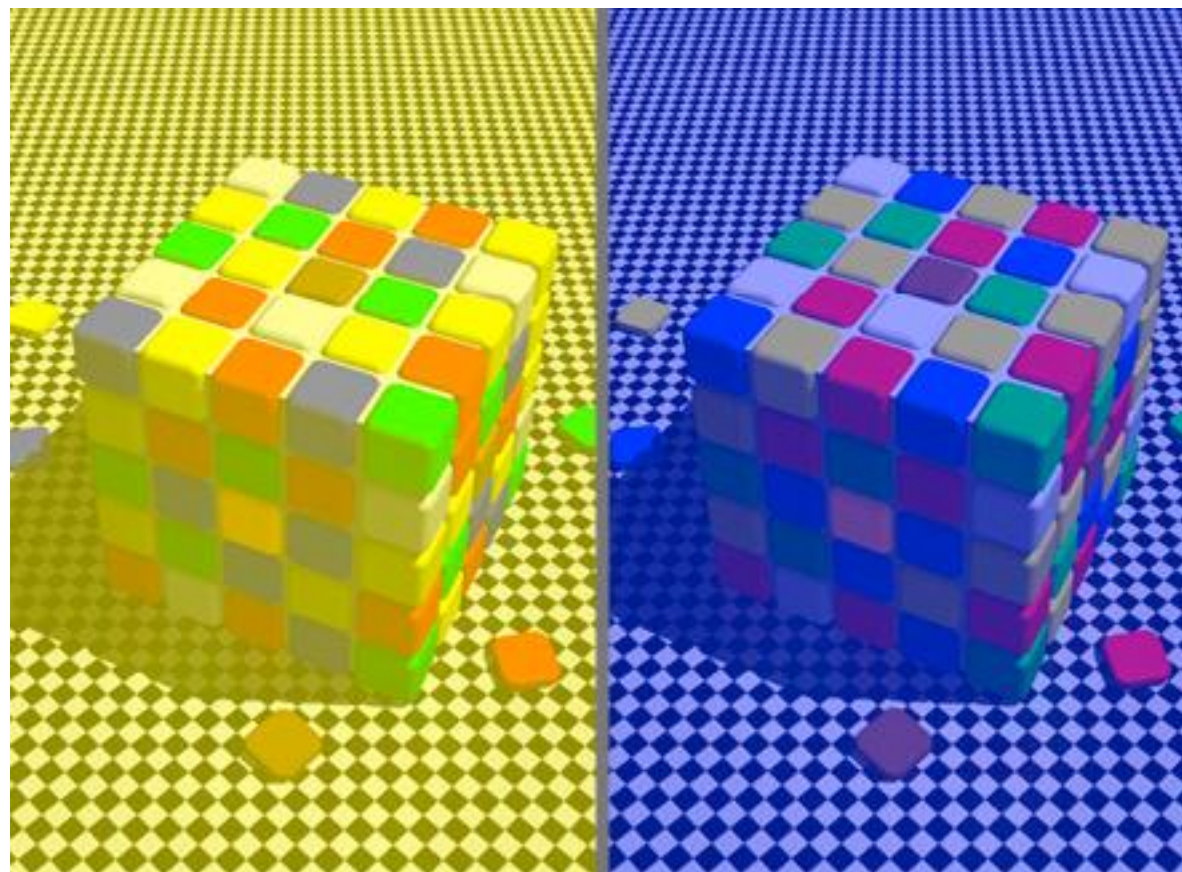


Edward H. Adelson



Relative colour judgements

- Color constancy across broad range of illumination conditions



Channels: Matching Groups & Rankings

➔ Magnitude Channels: Ordered Attributes

Position on common scale



Position on unaligned scale



Length (1D size)



Tilt/angle



Area (2D size)



Depth (3D position)



Color luminance



Color saturation



Curvature



Volume (3D size)



➔ Identity Channels: Categorical Attributes

Spatial region



Color hue



Motion



Shape



Best
Effectiveness
Least

Expressiveness principle:

- Match channel and data characteristics

Effectiveness principle:

- channels differ in accuracy of perception
- Encode most important attributes with highest rank channels

Channels: Matching Groups & Rankings

➔ Magnitude Channels: Ordered Attributes

Position on common scale



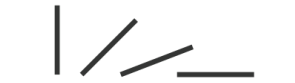
Position on unaligned scale



Length (1D size)



Tilt/angle



Area (2D size)



Depth (3D position)



Color luminance



Color saturation



Curvature



Volume (3D size)



➔ Identity Channels: Categorical Attributes

Spatial region



Color hue



Motion



Shape



Best
Effectiveness
Least

Same
Same

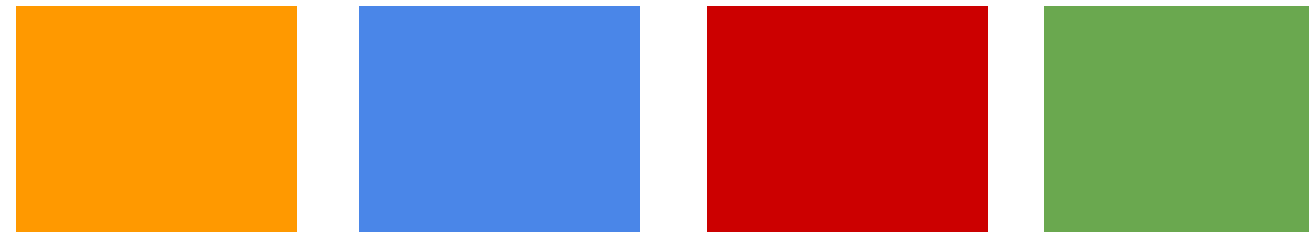
Effectiveness Ranking:

Note for both channel types spatial position are at the top, why?

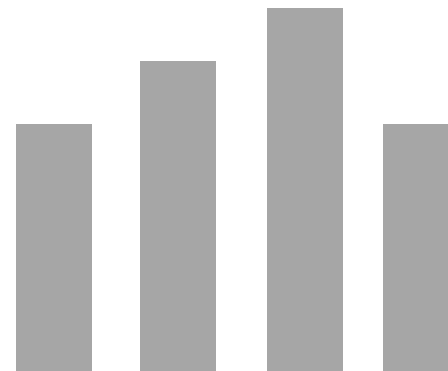
The choice of which attribute to encode with position is crucial in visual encoding

... because position coded attribute dominate user's mental model (internal mental reps used for thinking and reasoning)

Color Hue (identity channel) suitable for **categories**



Position (magnitude channel) suitable for **quantities**



Implications for design:

be mindful of ranking of visual variables. **Use highest ranked channels** for most important information whenever possible (with reason).

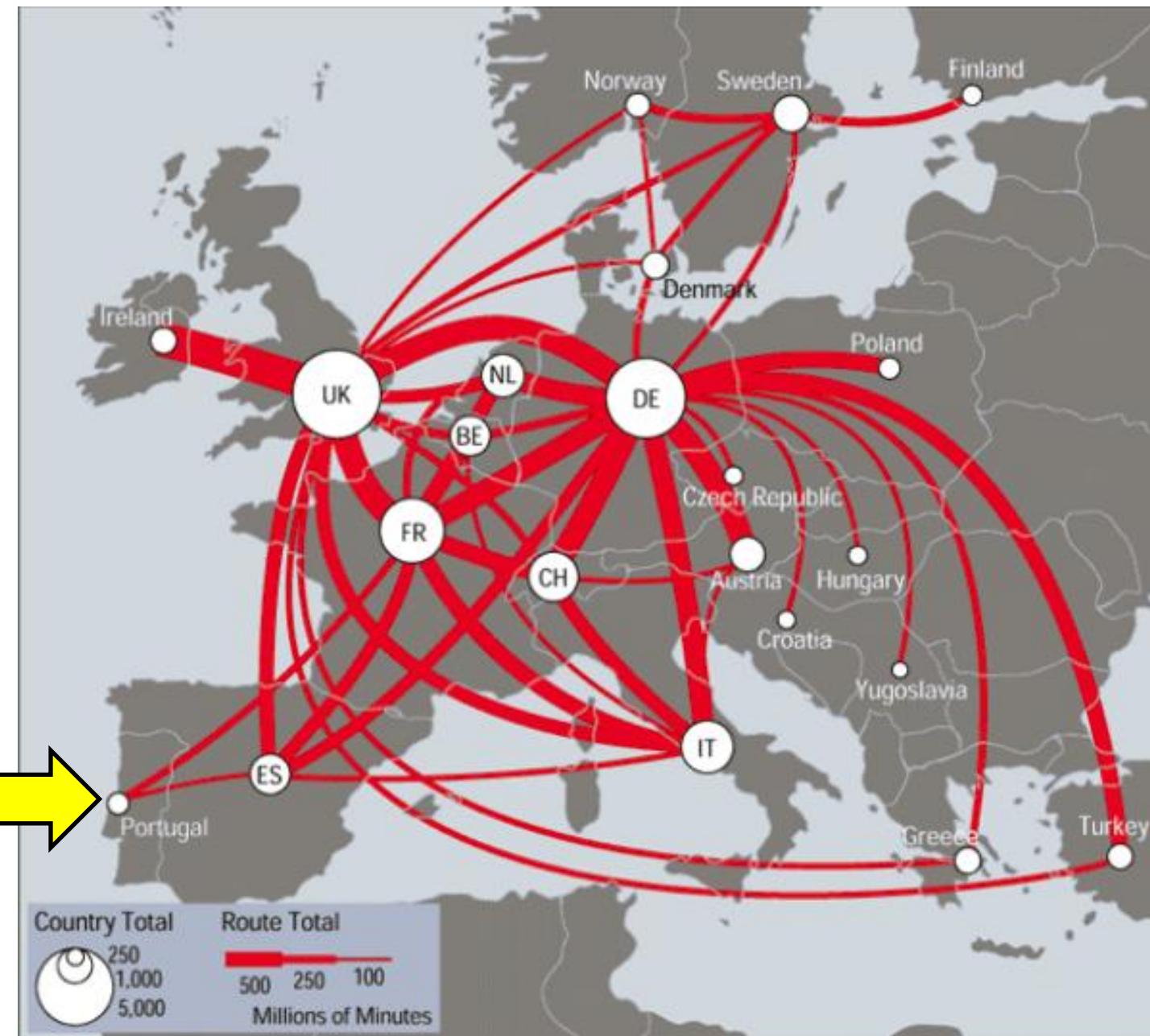
Keep in mind:

position and thus spatial layout is the “king” of visual channels. **Think about how you use space first.**

Discriminability: How many usable values?

- Visual channel ability to support sufficient number of attribute levels/values to show
- Linewidth has only limited number of discriminable bins i.e., okay for smaller number of attribute's values

Example of effective linewidth



Discriminability: How many usable values?

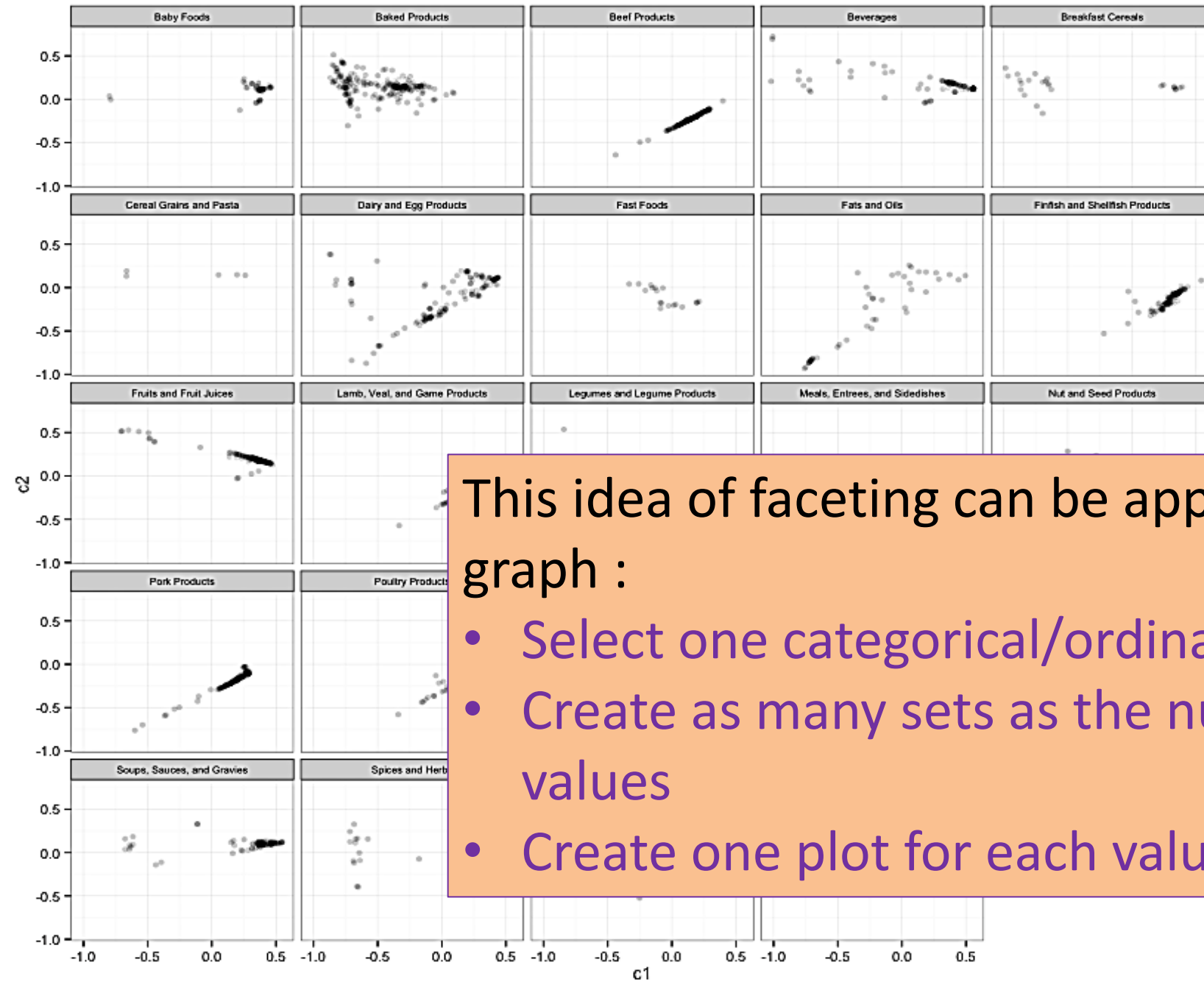
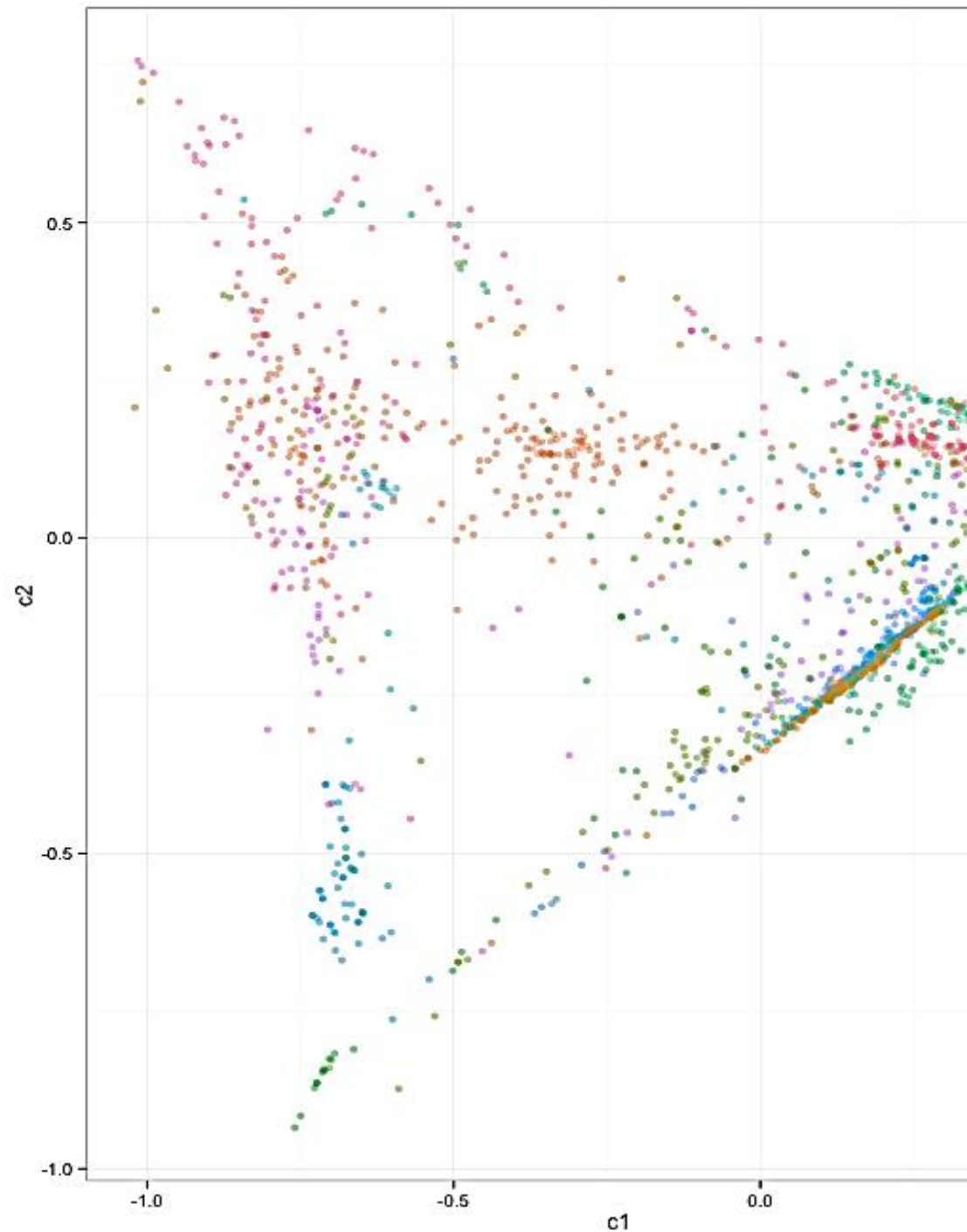
Implications for design:

Do not overestimate the number of available bins. For most visual channels the number is surprisingly low.

When you have **too many categories** find a way to **group/bin** the data further.

... or change channel! Example: you can often **trade-off color with space** ...

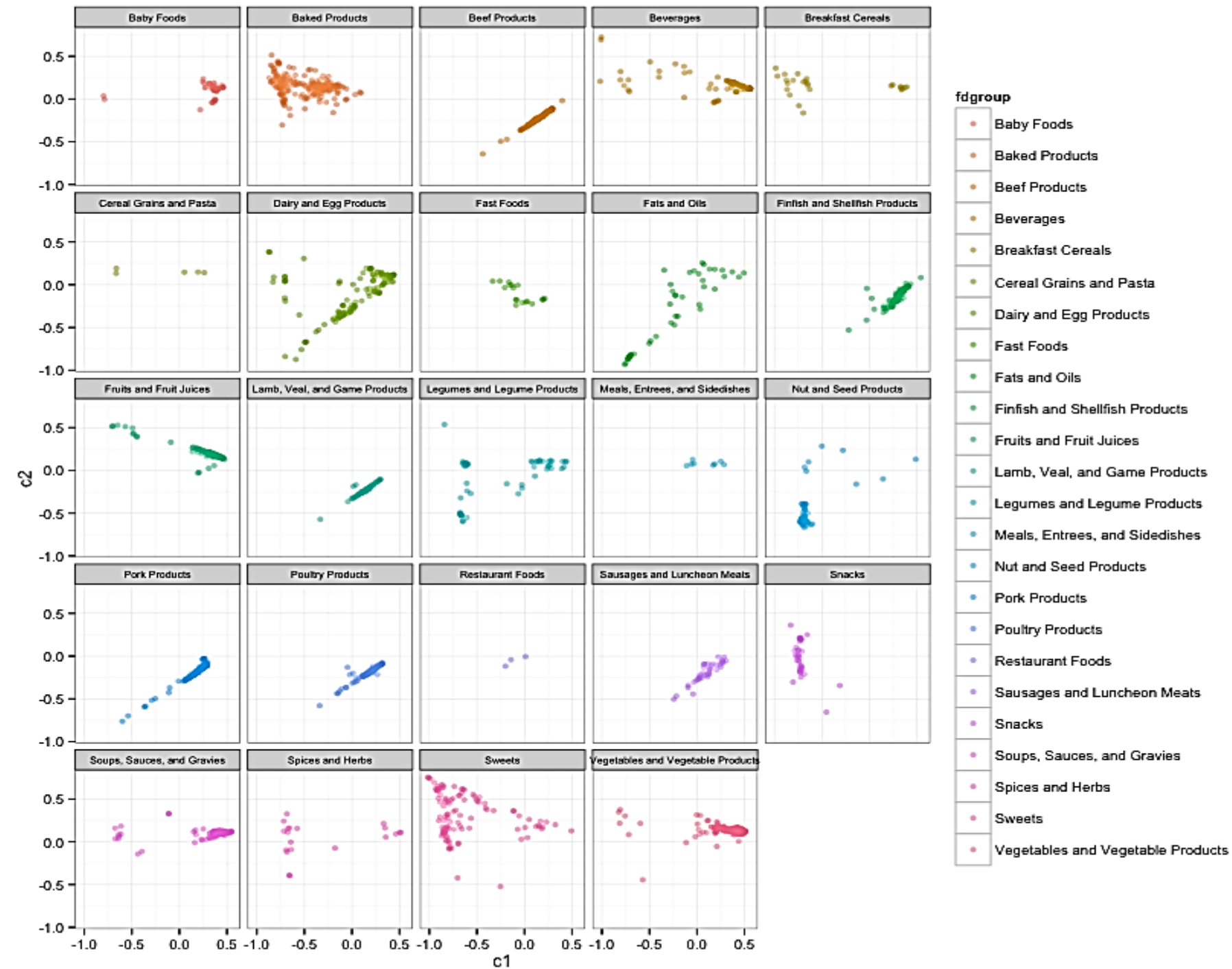
Change encoding



This idea of faceting can be applied to any graph :

- Select one categorical/ordinal attribute
- Create as many sets as the number of values
- Create one plot for each value

Change encoding: other option

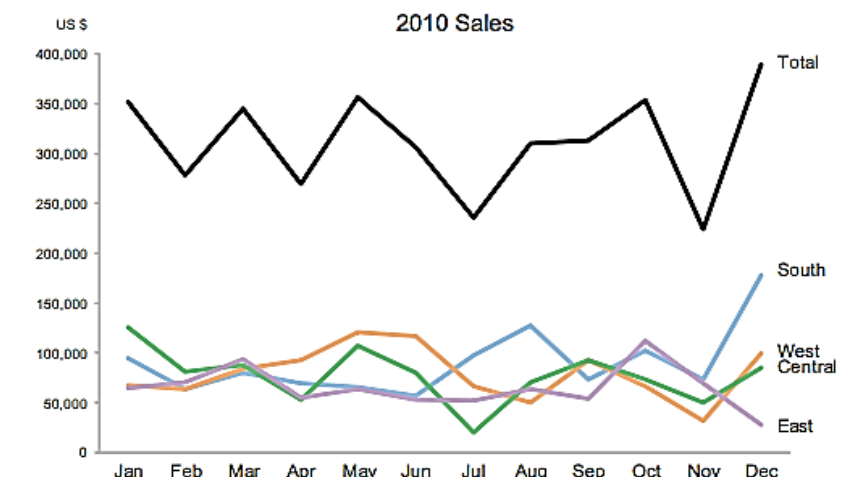
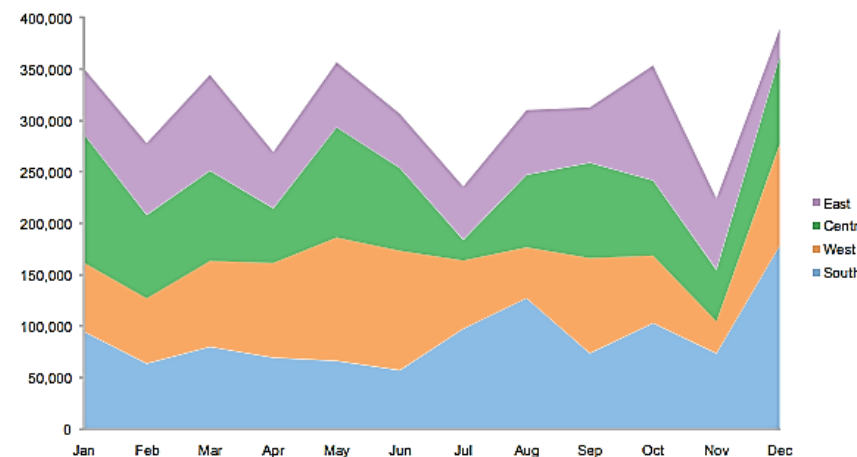
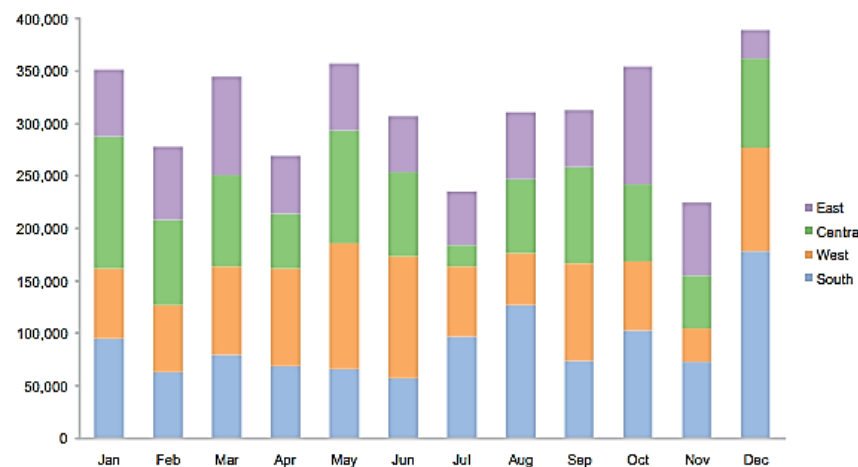
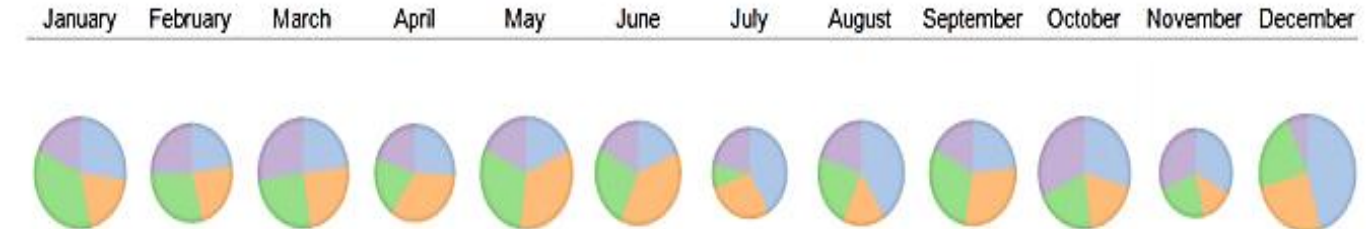


- Visual encodings → Marks & Channels
- Marks → map data item
- Channels → map attributes
- Channel principles (expressiveness & effectiveness)
- Channel effectiveness principles (four)

Class Activity (Effectiveness)

Which one is **more effective** to compare sale trends across regions?
Why?

Region	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
East	64,162	70,172	93,657	55,056	63,631	53,040	51,974	63,272	54,110	112,284	69,697	27,437
Central	125,392	80,851	87,645	52,996	107,159	80,349	19,907	70,431	92,568	72,961	49,881	85,084
West	67,364	63,742	83,856	92,412	120,284	116,618	66,692	49,671	92,920	65,971	31,516	99,000
South	94,572	63,234	79,491	68,963	65,868	56,659	97,101	126,879	73,240	102,589	73,044	177,943
Total	\$351,490	\$277,998	\$344,649	\$269,425	\$356,942	\$306,666	\$235,674	\$310,254	\$312,839	\$353,805	\$224,139	\$389,465

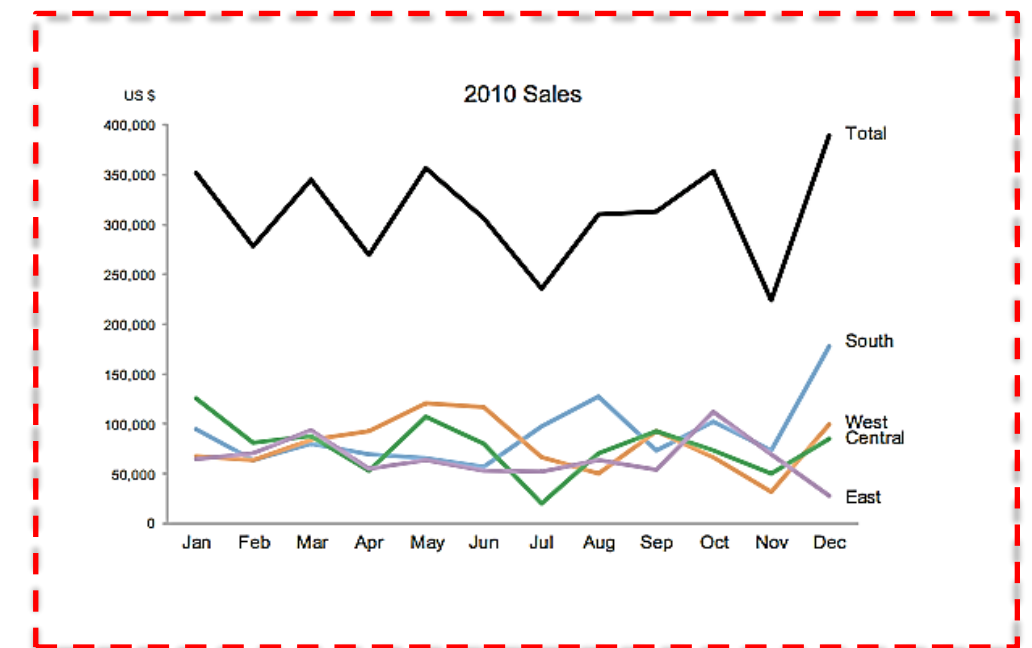
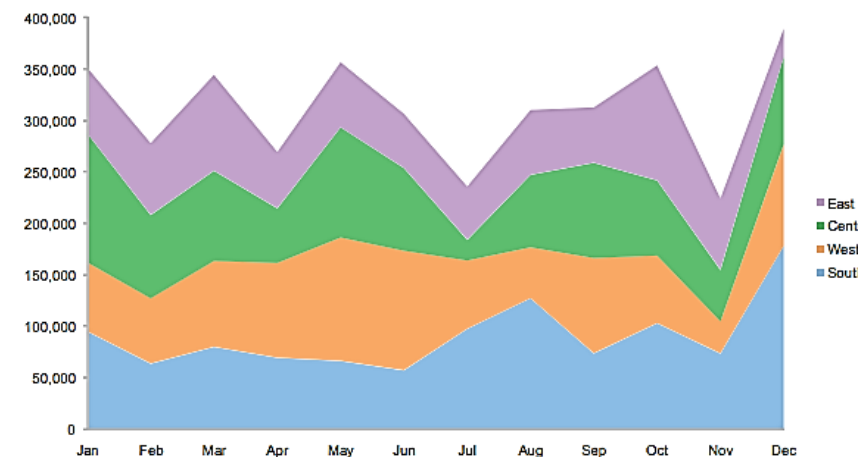
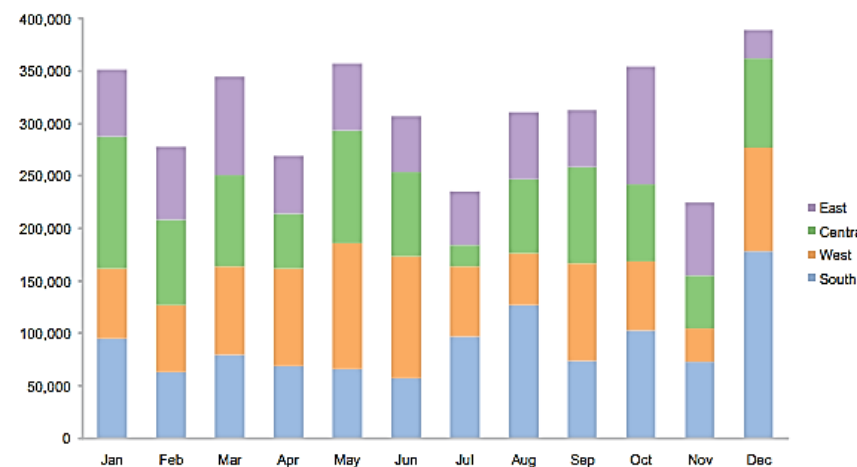
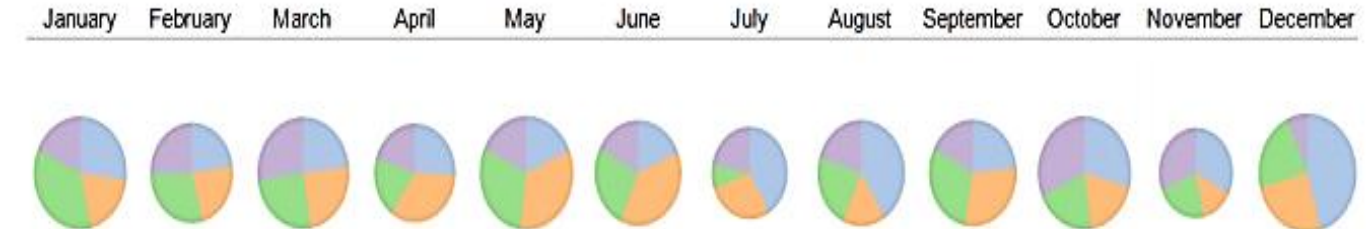


Example taken from Stephen Few: http://www.perceptualedge.com/articles/visual_business_intelligence/displays_for_combining_time-series_and_part-to-whole.pdf

Class Activity (Effectiveness)

Which one is **more effective** to compare sale trends across regions?
Why?

Region	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
East	64,162	70,172	93,657	55,056	63,631	53,040	51,974	63,272	54,110	112,284	69,697	27,437
Central	125,392	80,851	87,645	52,996	107,159	80,349	19,907	70,431	92,568	72,961	49,881	85,084
West	67,364	63,742	83,856	92,412	120,284	116,618	66,692	49,671	92,920	65,971	31,516	99,000
South	94,572	63,234	79,491	68,963	65,868	56,659	97,101	126,879	73,240	102,589	73,044	177,943
Total	\$351,490	\$277,998	\$344,649	\$269,425	\$356,942	\$306,666	\$235,674	\$310,254	\$312,839	\$353,805	\$224,139	\$389,465



Example taken from Stephen Few: http://www.perceptualedge.com/articles/visual_business_intelligence/displays_for_combining_time-series_and_part-to-whole.pdf